Embedded Spaces for Discovering New Classes

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Abstract

A representation can encode a novel class perfectly and still be useless for finding it: a supervised contrastive encoder reaches a holdout-vs-rest probe AUC of 0.94 while scoring zero at event-level detection. We study open-world class discovery as a two-sample problem and ask what makes an embedding *discoverable* rather than merely decodable. We give the pool an algebra — purity is bounded by $\min(1, b/q)$, a property of the cut alone — and derive a label-free cut from the Neyman–Pearson density ratio that already serves as the loss and the test statistic. Across a design cube of ten contrastive objectives, a leakage-free from-scratch CIFAR lineage, and three transfer datasets ordered by distance from pretraining, we find that the probe/discoverability dissociation is a many-class phenomenon (adding sliced-isotropy to SupCon is a probe and top-1 tie at 100 classes and lifts Euclidean novelty AUC $0.23 \rightarrow 0.62$; at 10 classes it buys nothing), that holdout draws dominate seeds by a factor of eight, and that the construction which survives every backbone and every draw is the simplest one: freeze the encoder and iterate anchors against a density-ratio pool ($143/144$ galaxy10 draw$\times$seed cells above the purity gate, where the fine-tuning loop on the same parents clears it on $0/5$ to $5/5$ draws by backbone). Taken to its limit — the pretrained trunk with no training at all — the same construction clears the gate on every galaxy10 and DTD draw and beats every fine-tuned objective there, sits at the gate on flowers, and is dead on aircraft and cars; the ordering is by distance of the novel class from the seen manifold, not by base rate. A residual decomposition survives a width-matched control and composes with discovery as a substitute more often than a complement. We report where the method fails: at a 1% novel rate the label-free cut declines on 17 of 20 clean spaces, and the limit is our estimator, not the representation.

1 Introduction

A representation can encode a novel class perfectly and still be useless for finding it. This is not a hypothetical failure mode; it is the ordinary situation. Train a supervised contrastive encoder on a set of known classes, hold one class out, and a linear probe trained to separate the held-out class from the rest reaches an AUC of 0.94. The same space, asked to flag novel examples without being told what to look for, scores exactly zero at the event level. The information is present and linearly accessible. It is simply not accessible to anyone who does not already have the answer.

The gap between those two numbers is the subject of this paper. It is the gap between a representation being *decodable* and being *discoverable*, and it matters because scientific discovery lives entirely on the second side of it. A search for a new particle, a new galaxy morphology, or a new failure mode of a deployed model does not come with labels for the thing being searched for. What it has is a large corpus, a smaller reference sample of what is already understood, and a question: is there something here that the reference does not explain?

Most of the representation-learning literature optimises for the first side. Linear probe accuracy, k-NN accuracy, and transfer benchmarks all measure how well a space supports a decision that someone else has already specified. They are the right metrics for that purpose. They are close to uninformative about whether a space will surface structure nobody has named yet, and — as the two numbers above indicate — they can be actively misleading, because the training objectives that maximise them are not the objectives that make distances mean anything in absolute terms.

What we do. We treat discovery as the target and work backwards. The pipeline we study is the standard open-world loop: score every point in a corpus against a reference of labelled examples, take the most anomalous ones as a *pool*, cluster the pool to propose new class anchors, and adapt the representation to the proposal. Every stage of that loop is downstream of one quantity — the fraction of the pool that is genuinely novel, which we call *pool purity*. Purity is the only forward-looking metric in the pipeline. Every other measure describes a space you already have; purity predicts whether the next step will improve it or corrupt it. Below a threshold the loop does not merely fail to help: it clusters background, plants anchors on it, and fine-tunes the representation to make that structure sharper.

Contributions.

1. **An algebra for the pool, and a ceiling.** Writing b for the novel base rate, q for the pool fraction, and ε_s for the fraction of novel points retained, pool purity is $\Pi = \varepsilon_s b/q$, so the enrichment Π/b is exactly ε_s/q and $\Pi \leq \min(1, b/q)$. The ceiling is a property of the cut alone. It says that a pool threshold inherited without derivation — as ours was — can cap achievable purity far below the gate no matter how good the representation is, and it separates "the space is bad" from "the cut is bad" for the first time.

2. **A label-free rule for the cut.** Because purity requires the novel labels, tuning the cut on purity is oracle tuning. We derive the novel-count estimator that the Neyman–Pearson critic already provides, and select the tightest cut that retains enough *estimated* novel points to remain clusterable. The rule uses only the corpus and the labelled reference. On synthetic data with an exact critic it recovers the oracle cut exactly; on real embeddings it raises pool purity from 0.232 to 0.778 where the scorer is strong, and refuses — reporting that it has refused — where it is not.

3. **A regime split, stated as a condition rather than a caveat.** Single-class novelty at a rate of $\sim$1% and multi-class novelty at $\sim$10% are different problems with different answers: which pooling statistic wins, whether stricter thresholds help or hurt, and whether the gate is reachable at all invert between them. We report them separately throughout.

4. **A reduced set of objectives, and the regime in which they differ.** From a design cube of ten contrastive objectives we identify the few that earn a place for discovery specifically, and show that the ranking by discovery is not the ranking by probe. On a from-scratch CIFAR-100 lineage, adding sliced-isotropy to a supervised contrastive objective is a tie on the linear probe (0.937 vs. 0.940) and a tie on supervised top-1 (0.597 vs. 0.608, inside the 0.017 noise floor), while raising every detection column — Euclidean AUC $0.233 \rightarrow 0.620$, tied-Mahalanobis $0.279 \rightarrow 0.602$. The same comparison on a from-scratch CIFAR-10 lineage goes the other way on *both* currencies (probe 0.884 vs. 0.905, Mahalanobis 0.454 vs. 0.552): the dissociation is a many-class phenomenon, and we state it as one. On the few-class lineage the objective that holds both currencies is supervised distance-NPLM with a class-wise marginal (probe 0.921, top-1 tied with SupCon, Euclidean AUC 0.729, per-event 0.19 against SupCon's 0.00).

5. **Two constructions, and a decomposition of what they buy.** We study a residual decomposition — train a second encoder on what the anchors fail to explain, then concatenate — and frozen-space discovery, in which only the anchors move. The residual's flagship gain survives a width-matched control (a same-width non-residual head recovers $+0.02$ of a $+0.15$ probe gain and loses geometry), and on clean CIFAR-10 it is a two-currency gain over its parent. Frozen-space discovery with a density-ratio pool is the one construction in the campaign that is both backbone- and holdout-independent: it clears the purity gate on $143/144$ galaxy10 draw$\times$seed cells (95% lower bound 0.96), on $8/8$ clean CIFAR-10 objectives, and on every random draw of DTD, where the fine-tuning loop on the same parents clears it on $0/5$ to $5/5$ draws depending on the backbone. It needs no training: the pretrained trunk alone, pooled by density ratio, reaches 0.56–0.60 purity on galaxy10 and 0.26 on DTD ($30/30$ draw cells), above every fine-tuned objective on both (Table 1). Because both constructions can be composed with the discovery loop, we report them in a 2×2 ablation (construction $\times$ discovery) rather than as a single delta. The composition is superadditive in 12 of 20 cells but concentrated: the effect that takes per-event detection from nominal to $6\times$ nominal lives in one cell,

while the two largest gains elsewhere turn out to be main effects of the construction and of discovery separately (Section 4). We state this narrowly because the decomposition is what makes it checkable.

6. **A leakage-free lineage for the claims that need one.** Every CIFAR cell built from hub-pretrained weights inherits an encoder that already saw the held-out class, which makes it unusable as evidence for discovery. We rebuild the CIFAR battery from random initialisation and report the pretrained cells only as an ablation. Dataset ordering throughout is by distance from pretraining rather than alphabetically, so the reader can see which claims rest on a backbone that could have seen the answer.

What this paper is not. We do not claim that a regulariser which makes embedded distance behave like a log-likelihood thereby improves discovery; we tested that directly and it does not hold (Section 6). We do not claim the pipeline works everywhere. A substantial part of what follows is a characterisation of where it does not, and why — in several cases the obstruction is the novel-class rate rather than anything about the representation, which is a limit no encoder can move.

2 Setup: discovery as a two-sample problem

The open-world protocol. We are given a corpus $\mathcal{D}$ of N embedded points, of which an unknown fraction b belong to classes never seen in training, and a reference $\mathcal{R} \subset \mathcal{D}$ of points carrying known class labels. No label for any novel class is available at any point — not for choosing hyperparameters, not for setting thresholds. We state this explicitly because the constraint is easy to violate accidentally, and because a method that quietly violates it reports numbers that cannot be reproduced in the situation it claims to address.

The loop. One round proceeds as follows. Score every point by an anomaly score $s(x)$; threshold to obtain a pool $\mathcal{P}$; cluster $\mathcal{P}$ and append the resulting centroids to the anchor set; pseudo-label every pooled point by its nearest discovered anchor; and adapt the representation (or, in the frozen variant of Section 4, only the anchors) to the enlarged anchor set. The round repeats with the updated anchors.

Pool purity, and why it gates everything. Let Π denote the fraction of $\mathcal{P}$ that is truly novel. The pool feeds directly into a clustering step with no supervision downstream of it, so $\mathcal{P}$ is the last point at which a mistake can still be corrected and nothing after it checks the work. At $\Pi \approx 0.01$ the clustering is not finding the novel class; it is partitioning background, and the subsequent fine-tune sharpens that partition. Empirically, in the multi-holdout regime, the probe drops by 0.031 to 0.038 wherever such pools form (under a single held-out class the same fine-tune is often probe-*positive* on weak trunks, because it acts as a calibration repair; Section 6). Purity therefore behaves as a gate rather than a gradient: above roughly 0.15 the loop compounds across rounds, and below it the loop is harmful. We report Π as the primary metric and treat the gate as a precondition for running the pipeline at all.

3 Algorithms

This section describes the objectives that produce the embedding, the statistic that scores it, and the rule that turns that statistic into a pool. The three are usually treated as separate literatures; a contribution of this work is that the middle one — the Neyman–Pearson density ratio — is the same object in all three roles, which is what makes the last one derivable rather than tuned.

3.1 Objectives

We work within a design cube spanning {instance, supervised} positives $\times$ {cosine, bilinear, distance} critics $\times$ {softmax, NPLM} estimators $\times$ {none, SIGReg, class-wise SIGReg} marginals, which contains SimCLR, SupCon, LeJEPA/VISReg and the density-ratio objectives as corners. Of the ten arms we ran, three earn a place for discovery.

SupCon + SIGReg (strong). Supervised positives, cosine critic, softmax estimator, and a sliced isotropic-Gaussian regulariser at weight $\lambda = 5$. This is the discovery workhorse: it produces the highest pool purities we measure. Two points of care. First, λ matters and is not a formality — the same objective at the library default $\lambda = 1$ is a different algorithm and its numbers must not be pooled with these (Appendix E). Second, we make no claim that SIGReg helps *because* it renders distance a log-likelihood; that mechanism is tested and rejected in Section 6. SIGReg earns its place empirically, and structurally: under an isotropic marginal Euclidean distance is Mahalanobis distance, which is what licenses the unit-variance clustering criterion used downstream.

Supervised distance-NPLM. Supervised positives, distance critic, and the NPLM estimator

$$\mathcal{L}[g] \;=\; \mathbb{E}_{\mathcal{R}}[e^{g} - 1] \;-\; \mathbb{E}_{+}[g]\,, \tag{1}$$

whose minimiser is the *absolute* log density ratio, fixed by the calibration $\mathbb{E}_{\mathcal{R}}[e^{g^\star}] = 1$ with no free additive constant. Unlike a softmax-normalised critic, this makes a single embedded distance a statement about relative likelihood rather than a statement only about rank. It is the strongest arm in the low-class-count regime and the most robust across pretraining backbones. Its class-wise-marginal variant (NPLM-dist. + classwise SIGReg) is the arm that holds both currencies on the leakage-free CIFAR-10 lineage and the only supervised arm that the discovery loop improves on the leakage-free CIFAR-100 lineage (Appendix B).

SupCon (plain). Supervised positives, cosine critic, softmax estimator, no marginal. Retained as the baseline and as the parent for the residual construction. It is the best space in the paper by linear probe and by semantic agreement, and one of the weakest by every calibration measure — which is the dissociation this paper is organised around, in a single arm.

We drop plain SimCLR, LeJEPA/SIGReg-SSL, instance NPLM-bilinear and instance NPLM-distance to a single ablation row: none reaches the Pareto front for discovery, and the bilinear critic in particular carries roughly three times the seed variance of the distance critic.

3.2 The Neyman–Pearson statistic, in three roles

Let $\mathcal{R}$ be the reference and $\mathcal{D}$ the corpus, with $w = N_{\mathcal{D}}/N_{\mathcal{R}}$. The empirical NP objective

$$\mathcal{L}[f] \;=\; w \sum_{x\in\mathcal{R}} \left(e^{f(x)} - 1\right) \;-\; \sum_{x\in\mathcal{D}} f(x) \tag{2}$$

has minimiser $f^\star(x) = \log\bigl(p_{\mathcal{D}}(x)/p_{\mathcal{R}}(x)\bigr)$, obtained by setting the functional derivative $p_{\mathcal{R}} e^{f} - p_{\mathcal{D}}$ to zero, and therefore satisfies

$$\mathbb{E}_{\mathcal{R}}\left[e^{f^\star}\right] = \int p_{\mathcal{R}} \cdot \frac{p_{\mathcal{D}}}{p_{\mathcal{R}}} = 1. \tag{3}$$

Equation 2 is Equation 1 with pairs replaced by events. The same functional appears as a contrastive *loss* over pairs, as a two-sample *test statistic* $t_{NP} = -2\mathcal{L}$ distributed as a Wilks log-likelihood ratio, and — below — as a *pool scorer*. We parameterise f as a small kernel ensemble with learnable centres and amplitudes and anneal the bandwidth from the median pairwise distance downwards; a fixed bandwidth is calibrated only for one embedding scale and goes blind when the geometry is rescaled.

Equation 3 is also a free diagnostic, and we report it. A fitted critic need not satisfy it, and a value far from 1 means the numbers built from that fit are unreliable. Measured on trained spaces it is 0.997 to 1.019 in-sample; out-of-sample it ranges to 2.7, reflecting mild overfitting of the reference subsample.

Why the ratio and not the distance. Distance-based pooling asks which points are far from everything. The density ratio asks which points are over-represented relative to what the reference explains. These come apart, and the difference is decisive: a novel class is not a scatter of far-flung outliers but a *dense clump* sitting where the reference has no mass. On a 100-class benchmark the extreme distance tail is owned by background outliers, and tightening the distance threshold makes purity *worse* ($0.036 \to 0.002$), whereas tightening the ratio threshold makes it better ($0.214 \to 0.268$ with the encoder exactly frozen; an earlier archive that let BatchNorm statistics drift read $0.232 \to 0.358$, and we do not use it). Same space, same rate, opposite response.

3.3 Choosing the pool: an algebra and a rule

The algebra. Let $q = |\mathcal{P}|/N$ and let $\varepsilon_s, \varepsilon_b$ be the fractions of novel and background points retained. Then $q = \varepsilon_s b + \varepsilon_b(1-b)$ and

$$\Pi \;=\; \frac{\varepsilon_s\, b}{q}, \qquad E \;\equiv\; \frac{\Pi}{b} \;=\; \frac{\varepsilon_s}{q}, \qquad \Pi \;\le\; \min\Big(1,\; \frac{b}{q}\Big). \tag{4}$$

Three consequences. The enrichment is exactly the lift at operating point q. Purity increases monotonically as q shrinks, so there is no interior optimum and targeting a purity *level* is the wrong objective. And the ceiling $\min(1, b/q)$ is a property of the cut alone: at $q = 0.05$ and $b = 0.01$ no scorer, however good, can exceed $\Pi = 0.20$. A threshold chosen without reference to b can therefore cap purity near the gate independently of the representation — and the widely used "95th percentile of the reference" is such a threshold.

The rule. Purity cannot be used to choose q, since computing it requires the novel labels. The critic supplies the missing quantity. Under the contamination model $p_{\mathcal{D}} = (1-b)p_{\mathcal{R}} + b\,p_S$, with $r = e^f$, the posterior that a corpus point is novel is $1 - (1-b)/r$; dropping the unknown b and taking the positive part gives the label-free weight

$$w(x) \;=\; \left[1 - \frac{1}{r(x)}\right]_+, \qquad \mathbb{E}_{\mathcal{D}}[w] \;=\; \mathrm{TV}(p_{\mathcal{D}}, p_{\mathcal{R}}) \;=\; b\,\mathrm{TV}(p_S, p_{\mathcal{R}}) \;\le\; b, \tag{5}$$

so $\sum_{x\in S} w(x)$ estimates how many novel points a subset S contains, exactly when the novel class is disjoint from the reference and biased *downward* under overlap — a safe direction, since it tightens the cut. Since purity rises as q falls, the binding constraint is not purity but whether a cluster survives, and that is what we target: ordering the corpus by f and writing $\widehat{N}(k)$ for the null-corrected sum of w over the top k points,

$$k^\star \;=\; \min\big\{\, k : \widehat{N}(k) \ge n_{\min} \,\big\}, \qquad q^\star = k^\star/N, \qquad k_{\max} = \Big\lfloor \widehat{N}(N)\,/\,n_{\min} \Big\rfloor. \tag{6}$$

The same weights bound the clustering search range, replacing a bound that in the original pipeline was set from the number of held-out classes — a quantity an open-world method does not have.

The null correction, which is not optional. Equation 5 is a positive part, so *any* spread in a fitted f manufactures apparent novelty. On pure-noise scores with no novel class at all, the raw sum returns 665 "novel" points out of 8000 — enough for the rule to fire on nothing and hand the loop a pool of pure background. Subtracting the mean weight of the reference does not repair this, because top-ranked null points carry far above-average weight and their tail survives subtraction. We instead subtract a *rank-matched* null: under the null hypothesis the top q of the corpus is distributed as the top q of the reference, so $\widehat{N}(k) = \sum_{i\le k} w(x_{(i)}) - k \cdot \overline{w}_{\mathcal{R}}^{(q)}$, with $\overline{w}_{\mathcal{R}}^{(q)}$ the mean weight of the top-q fraction of the reference. This cancels to zero under the null while preserving a real excess. When $\widehat{N}(N) < n_{\min}$ the rule declines to pool and reports that it has declined; that refusal is an output of the method, and we report it as one.

Constants. $n_{\min}$ and the clipping range on q are set once, globally, from a sweep on held-out embeddings rather than inherited: purity falls monotonically as $n_{\min}$ grows, so it is set as small as the clustering will bear.

4 Constructions

Both constructions in this section are defined *relative to an anchor set*, which is what lets them compose with the discovery loop: discovery enlarges and corrects the anchors, so a construction defined against them is a different construction after discovery than before. That composition is the part of this work with no counterpart we are aware of, and it is also the part most easily overstated, so we measure it as an interaction rather than a delta.

4.1 The residual decomposition

Let μ_y be the anchor for class y and z an embedding. The residual $r = z - \mu_y$ is what the anchor set fails to explain about z. We train a second encoder on r under the same objective family as the parent and either use it alone (*residual*) or concatenate it with the parent (*concat*).

The motivation is that a supervised objective is free to discard any variation that does not separate the classes it was given, and novelty is precisely variation of that kind. The residual target makes that discarded structure the only thing the child can see.

On its own, the residual buys detection at no accuracy cost. On the from-scratch CIFAR-100 lineage, concatenating a residual onto a supervised contrastive parent moves Euclidean novelty AUC from 0.233 to 0.521 while the linear probe moves by $+0.0015$, a gain two orders of magnitude smaller than the detection gain and of the same order as the probe's own seed spread ($\sigma = 0.0008$). This is the pattern across cells: the pure residual usually *loses* probe accuracy (four of six strong-parent cells, -0.037 to -0.077), and the concatenation recovers it. We therefore report concat as the construction and the bare residual as a diagnostic.

The few-class lineage tells the complementary half. On from-scratch CIFAR-10 the concatenation is a real gain on *both* currencies over its SupCon parent (probe $+0.034$, Euclidean $+0.056$, tied-Mahalanobis $+0.015$, top-1 tie), and the SupCon+SIGReg parent's NPLM-residual child is the best-calibrated clean CIFAR-10 space we measure (Euclidean AUC 0.758, per-event 0.12 against the parent's 0.01). The residual buys probe where the parent already has geometry, and geometry where the parent already has the probe (Table 21).

The gain is not width. The concatenation is two full encoders, so a referee should ask whether a single encoder of the same width would do as well. It does not: on cars/VISReg a 200-D SupCon head recovers $+0.02$ of the residual concat's $+0.15$ probe gain and *loses* Euclidean AUC (0.675 vs. the parent's 0.715) where the concat gains it (0.747); the concat beats the control by $+0.12$ probe, $+0.08$ Euclidean and $+0.11$ Mahalanobis, each five times the noise floor. On galaxy10 the probe gain is real on LeJEPA ($+0.06$ over the control) and a tie on DINO ($+0.01$), and on both backbones the wider single head is the better *geometry* space (Table 13). Width buys calibration cheaply; the residual buys the probe.

4.2 Composing with discovery: a 2×2

Because the residual is defined against the anchors, it can be built before or after the discovery loop. Writing A for the parent alone, B for the parent after discovery, C for the construction before discovery, and D for both, the quantity of interest is not $D - A$ but the interaction

$$\Delta_{\text{int}} \;=\; (D - C) - (B - A), \tag{7}$$

which is zero for an additive stack and positive only when the two ingredients do more together than apart. We report it for per-event power at a nominal false-positive rate $\alpha = 0.05$, because that metric is bounded below by α: a parent at $A \approx \alpha$ has *no* ability to flag an individual novel sample, so a positive interaction there is a threshold crossing rather than an improvement.

The interaction is real, and it is narrow. On DTD with a DINO backbone, the residual construction gives $A = 0.040$, $B = 0.068$, $C = 0.023$, $D = 0.253$. Discovery alone is worth $+0.028$; the construction alone is worth -0.018 — it actively hurts — and composed they reach $6\times$ nominal, an interaction of $+0.203$. The concatenation behaves the same way ($+0.163$). Neither ingredient does anything alone. This is the strongest evidence we have that discovery and representation construction are not independent improvements.

Two apparent replications are not replications. The same protocol on galaxy10 produces the two largest raw gains in the study, and the decomposition shows that neither is composition. With a DINO backbone the residual takes per-event power from 0.021 to 0.299, but $C = 0.281$ already: the construction does this *by itself*, and the interaction is $+0.003$. With a LeJEPA backbone the gain from 0.007 to 0.206 is almost entirely the discovery main effect ($+0.190$), with an interaction of $+0.008$ — and it occurs in a cell whose round-2 pool purity is exactly zero. That last case is worth stating plainly, because we had written it in advance as a falsifier: if the composition helps where purity is zero, then the mechanism we proposed for it — that discovery supplies a richer and *correct* anchor set, against which the residual becomes informative — cannot be what is happening there.

Building the residual *after* discovery makes the mechanism visible. Across five cells the after-discovery concat beats the before-discovery one on four (DTD/DINO: probe $0.845 \to 0.898$, Euclidean $+0.10$, per-event $0.05 \to 0.24$; galaxy10/VISReg: $+0.01$, $+0.13$, $+0.17$; Table 14), and is a no-op on the fifth. But in every cell, including the two whose round-1 pool purity was 0.06, the images whose nearest anchor is a *discovered* one are 100% novel. The clustering step is noisy; the post-hoc nearest-anchor assignment is not. So the purity gate governs whether the *loop* compounds, not whether the anchors it planted sit on the novel class — which is why a construction defined against those anchors can help below the gate, and why the LeJEPA cell above is not a counterexample to the anchors being right, only to the loop having needed them.

What we therefore claim. Across five cells, two constructions, and two variants, 12 of 20 interactions are positive, but most are small and the threshold crossings driven by interaction rather than by a main effect occur in a single cell with a single construction. We do not claim superadditivity as a general property. The claim the data supports is weaker and still useful: *discovery and the residual are substitutes more often than complements.* Whichever of the two a cell is missing, one of them supplies it — galaxy10/LeJEPA needed discovery, galaxy10/DINO needed the residual, DTD/DINO needed both — and a practitioner who runs only one of them will, on most cells, capture most of the available gain. Every number in this subsection is a single seed; the interaction is a difference of differences and inherits the variance of both its terms, so we treat it as a finding to be replicated rather than a headline.

4.3 Frozen-space discovery

Freeze the encoder, score the corpus by the density ratio of Section 3.2, and iterate only the anchors. Nothing in the representation moves; the loop's only output is an enlarged anchor set. This is the construction the campaign's single-holdout results converge on, for a reason that only the draw-resampled tables make visible.

The fine-tuning loop is backbone-dependent. On galaxy10 — the one dataset none of our backbones has seen — the standard loop (encoder trains, distance pool) clears the purity gate on random single-holdout draws for $0/5$ classes with a DINO backbone, $2/5$ with LeJEPA and $5/5$ with VISReg (Table 38). The archived alphabetical draw had reported purities of 0.46–0.50 on every backbone; across five random classes the same arm is 0.13–0.19. Whether "discovery works on galaxy10" depends on which galaxy morphology is novel and which backbone embeds it, and no single-draw table can show that.

The frozen density-ratio pool is not. With the same parents frozen and the pool scored by the NP ratio, round-1 purity is 0.40–0.57 on every backbone, every objective and every held-out class, with no round-2 collapse: 143 of 144 draw$\times$seed cells clear the gate (Table 5), and the one miss is at 0.10. On the leakage-free CIFAR-10 lineage the same construction clears it on $8/8$ objectives (0.26–0.55) where the distance pool manages $3/8$ (Table 20). It costs nothing in the probe — the encoder does not move — and it removes the round-2 collapse that every fine-tuning run in Tables 38 and 41 shows: the collapse is the fine-tune destroying the pool, not the anchors. The construction does not even need the fine-tune: pooling the *pretrained* trunk's features, with seen-class centroids as anchors and no training at all, clears the gate on $30/30$ single-holdout draw cells on galaxy10 and DTD — at 0.56–0.60 on galaxy10, above the frozen fine-tuned head (0.40–0.57) and far above the fine-tuning loop (0.13–0.29), and at 0.26 on DTD, above the fine-tuned workhorse (0.22–0.24). Fine-tuning on the seen classes buys the probe and the multi-holdout regime; it does not buy single-holdout discoverability, and on galaxy10 it costs some. Where the base rate is low the same construction is marginal rather than dead — flowers at $1/102$ sits at 0.16–0.18, at the gate on 12 of 15 draws — and where the novel class sits inside the seen manifold it is dead regardless of rate (aircraft at $1/100$: 0.01; cars at $1/196$: 0.01; Table 15). That ordering — galaxy10, DTD, flowers, aircraft, cars — is the regime boundary of Section 5 stated on one construction.

What "frozen" has to mean. A trunk with BatchNorm keeps adapting its running statistics in training mode even with every weight frozen, and on the CIFAR ResNet those statistics absorbed the discovery corpus, novel points included. Setting the trunk to evaluation mode changes the numbers: the round-2 rise of our earlier frozen archive ($0.358 \rightarrow 0.418$) disappears ($0.268 \rightarrow 0.237$), and a paired A/B on the same checkpoint attributes the round-2 rise entirely to the statistics adapting ($0.237 \rightarrow 0.299$ with adaptation, $\sim$30% embedding drift, no probe cost). That is a legitimate construction — corpus-adaptive normalisation uses no labels — and we name it as one; but it is not "frozen", and its transfer to LayerNorm backbones is a null result (-0.006 round-2 purity across 45 galaxy10 draw cells). The frozen numbers in this paper are evaluation-mode numbers throughout.

5 Experiments

This campaign is reported while incomplete. Every table below states its own coverage, blanks are shown as "–" rather than dropped, and we mark which claims rest on a single seed. Datasets are ordered by *distance from pretraining* rather than alphabetically, because that ordering is what decides whether a result is evidence for discovery at all: galaxy10 was never seen by any backbone we use, DTD is task-novel, cars/flowers/aircraft are fine-grained but in-domain, and the CIFAR cells exist in two lineages — a from-scratch one that we report, and a hub-pretrained one whose encoder already saw the held-out class and which we therefore treat as an ablation only.

5.1 Objectives: the probe and the geometry come apart

Table 3 is the paper's central measurement, on the lineage where it is safe to make it. Adding sliced-isotropy to a supervised contrastive objective is a tie on the linear probe and a tie on top-1 accuracy — and it more than doubles every discoverability column. A reader who ranked these eight objectives by probe accuracy would pick the one that scores 0.00 on per-event detection.

That is the dissociation stated at the start of the paper, measured on an encoder that was trained from random initialisation and never saw the class it is asked to find. It is also a *many-class* measurement. The same eight objectives trained from scratch on CIFAR-10 (Table 18) rank differently: there SupCon's own geometry is the better one, the isotropy term buys nothing on either currency, and the objective that carries both is the class-wise-marginal distance-NPLM. Three datasets now agree on the shape: SIGReg earns the second currency where classes are many and near, and is neutral-to-harmful where they are few and separable.

5.2 Stability: draws matter, seeds do not

Two variance questions decide how much of this campaign is quotable.

The first is the holdout *draw* — which class is held out. Table 4 reports five draws per cell. Pool purity is stable across draws for the best objective (sd 0.007–0.014) while the linear probe over the same draws swings by more than 0.1, so purity is the better-conditioned measurement of the two, not merely the more relevant one.

The second is the *seed*. We pre-registered this as a falsifier: if seed-to-seed spread were comparable to draw-to-draw spread, every single-seed number in this paper would need a caveat. Table 5 settles it on 144 cells — $\mathrm{sd}_{\mathrm{seed}} = 0.021$ against $\mathrm{sd}_{\mathrm{draw}} = 0.174$, a factor of 8. The caveat that survives is about which class you hold out, not which seed you ran. We attach it to every single-seed result rather than dropping it.

Table 4 also carries a result we did not anticipate. On DTD only one objective clears the gate on every backbone: supervised distance-NPLM clears $5/5$ draws on LeJEPA and $0/5$ on DINO, while SupCon+SIGReg clears $15/15$ across DINO, LeJEPA and VISReg (Table 41). Base-independence is a stronger and more useful property than winning on average, and it is not visible in any single-backbone table. The single-holdout DTD purity is 0.22–0.24; the multi-holdout figure the same arm posts on the same dataset is 0.80, and the two are different regimes rather than a discrepancy.

The archived draw was the favourable one. Every headline number this campaign reported before it resampled holdouts was measured on the alphabetically last class. On galaxy10 that class sits at or above the *top* of the random-draw range for tied-Mahalanobis on 17 of 18 (objective, backbone) pairs and for purity on the arm that carried the claim (Table 38). We do not think this was a selection effect — the draw was fixed before any result existed — but a reader comparing this paper with our earlier technical report will find the earlier numbers systematically higher, and this is why.

dataset	rate	backbone	purity	$\geq$gate	fine-tuned loop, best objective
galaxy10	1/10	dino	0.598±0.061	5/5	0.131 (nplm-sup-ft)
		lejepa	0.591±0.040	5/5	0.190 (nplm-sup-ft)
		visreg	0.556±0.026	5/5	0.290 (supcon-ft)
dtd	1/47	dino	0.257±0.016	5/5	0.219 (ss-ft)
		lejepa	0.261±0.014	5/5	0.225 (ss-ft)
		visreg	0.267±0.014	5/5	0.241 (ss-ft)
flowers	1/102	dino	0.158±0.006	5/5	–
		lejepa	0.177±0.039	4/5	–
		visreg	0.159±0.016	3/5	–
aircraft	1/100	dino	0.016±0.010	0/5	–
		lejepa	0.010±0.007	0/5	–
		visreg	0.010±0.008	0/5	–
cars	1/196	dino	0.009±0.008	0/5	–
		lejepa	0.007±0.008	0/5	–
		visreg	0.010±0.006	0/5	–

Table 1: The regime boundary on one construction that needs no training. Round-1 pool purity of the pretrained trunk pooled by density ratio, single holdout, mean ± sd over five held-out classes. galaxy10 and DTD clear the gate on every cell and beat the best fine-tuned objective; flowers sits at the gate; aircraft and cars are dead although aircraft's rate equals flowers'. *Coverage:* 75 cells; pretrained ViT-B/16 features, identity head, anchors = seen-class centroids, density-ratio pool, anchors-only iteration, round 1; five single-holdout draws per (dataset, backbone); gate $= 0.15$. Reference = the best objective of the exp-70 fine-tuning loop on the same draws.

5.3 Constructions

Table 6 gives the 2×2 of Section 4 in full. The interaction is large in one cell, positive but small in most, and the two largest *totals* in the table are main effects rather than composition. We report all twenty rows rather than the favourable ones.

5.4 The regime boundary, on one construction

Table 1 runs the zero-training construction of Section 4.3 — pretrained features, seen-class centroids, density-ratio pool — on every transfer dataset at single holdout, five draws each. It orders the datasets more sharply than any fine-tuned table: galaxy10 (0.56–0.60) and DTD (0.26) clear the gate on all 30 cells and beat the best fine-tuned objective on the same draws; flowers sits at the gate $(0.16–0.18, 12/15)$; aircraft and cars are dead (0.01). Flowers and aircraft have the same novel rate, so the rate is not what separates them: the pretrained trunk already places an unseen flower species away from the seen ones and an unseen aircraft variant among them. The base rate is necessary — cars at $1/196$ is dead in both regimes — but the boundary is set jointly by the rate and by where the novel class sits in the pretrained space, which is what the earlier on-/off-manifold split measured with local intrinsic dimension. Two consequences for practice: on the datasets where discovery works, fine-tuning on the seen classes is not what makes it work (on galaxy10 it lowers single-holdout purity from 0.60 to 0.13–0.29 under the loop); and whether a new dataset is discoverable can be read off the pretrained trunk before any training is done.

5.5 Against the generalized-category-discovery benchmarks

The nearest established benchmark is generalized category discovery (GCD): DINO ViT-B/16, the first N classes known with half of their images labelled, and clustering accuracy under a single Hungarian assignment over all K classes on the unlabelled train set, with K given. We ran our constructions under that protocol (Table 2); our k-means row reproduces the published one on all four datasets to within 0.5 points, so the rest of the table is on the same footing.

Two things it shows. First, our objectives buy this metric only where the known set is rich: a SupCon head trained on the labelled images alone doubles k-means on Cars and Aircraft (two-thirds of GCD's Old accuracy, matching UNO+ on New) and *collapses* CIFAR-10 $(83.5 \to 61)$ — with five known classes and no unlabelled term, a supervised projection discards the structure the novel classes live in, which is precisely why GCD adds an unsupervised contrastive term on the unlabelled data. Our objectives never see the unlabelled set during representation learning; our discovery loop is what consumes it. Second, the density-ratio construction is the wrong tool for this metric, and the mismatch is informative: its pool is pure (0.70 on Cars) but BIC returns two or three anchors where the benchmark wants up to 98 novel clusters, so the K-free row posts the highest Old accuracy on CIFAR-100 (70.9, it protects the known classes) and the lowest New. Our pipeline locates novelty; the GCD metric scores partitioning it with K given. The fine-tuned rows (labelled data only, our exp-70 recipe) place the representation step of our objectives against theirs; the remaining asymmetry — their 200 epochs on labelled and unlabelled data versus our 20 on labelled — is the ablation a reader should want next, and it is one term in the loss.

method	CIFAR-10	CIFAR-100	Cars	Aircraft
frozen DINO trunk (ours, exp 144)				
k-means, raw features	83.5/86.1/82.2	51.9/52.4/50.8	12.4/10.7/13.3	16.0/8.9/19.5
ss-k-means, raw features	74.3/62.5/80.2	60.5/61.7/58.1	12.2/10.2/13.3	14.9/7.0/18.8
np-anchors, raw features (K free)	58.6/38.7/68.5	55.7/70.9/25.4	10.8/14.3/9.0	10.6/9.4/11.2
ss-k-means, SupCon head	52.4/49.6/53.8	46.4/57.9/23.3	27.1/44.8/18.3	29.4/39.8/24.2
ss-k-means, SupCon+SIGReg head	59.1/61.0/58.2	56.1/66.7/34.7	21.0/27.6/17.7	24.0/25.6/23.2
fine-tuned trunk, labelled data only (ours, exp 145)				
ss-k-means, SupCon trunk	–	–	–	–
ss-k-means, SupCon+SIGReg trunk	–	–	–	–
ss-k-means, NPLM-dist. trunk	–	–	–	–
published, fine-tuned on labelled + unlabelled				
k-means (paper)	83.6/85.7/82.5	52.0/52.2/50.8	12.8/10.6/13.8	16.0/14.4/16.8
GCD	91.5/97.9/88.2	73.0/76.2/66.5	39.0/57.6/29.9	45.0/41.1/46.9
UNO+	68.6/98.3/53.8	69.5/80.6/47.2	35.5/70.5/18.6	40.3/56.4/32.2
SimGCD	97.1/95.1/98.1	80.1/81.2/77.8	53.8/71.9/45.0	54.2/59.1/51.8
RLCD	97.4/96.4/97.9	83.4/84.2/81.9	64.9/79.3/58.0	60.6/62.2/59.8

Table 2: Under the Generalized Category Discovery protocol. Clustering accuracy All/Old/New. Our frozen-trunk k-means row reproduces the published one; a head trained on labelled data alone doubles it on the fine-grained sets and collapses novel structure on CIFAR-10. *Coverage:* 4 result files; GCD protocol (first-N known classes, 50% labelled, Hungarian ACC on the unlabelled train set, K known), DINO ViT-B/16, 3 split seeds (means shown).

objective	sup.	probe	top-1	acc	eucl	mahaT	per-ev
SimCLR	n	0.799	0.292	0.242	0.443	0.385	0.01
VISReg/LeJEPA	n	0.797	0.239	0.190	0.499	0.382	0.03
NPLM (unsup.)	n	0.757	0.223	0.181	0.413	0.361	0.01
SupCon	y	0.940	0.608	0.545	0.233	0.279	0.00
SupCon+SIGReg (repulse)	y	0.823	0.360	0.346	0.340	0.320	0.00
NPLM-dist.+classwise SIGReg	y	0.898	0.507	0.463	0.525	0.411	0.03
SupCon+SIGReg (λ **=5)**	y	0.937	0.597	0.588	0.620	0.602	0.07
NPLM-dist. (sup.)	y	0.820	0.254	0.232	0.408	0.345	0.05

Table 3: Objectives on the leakage-free CIFAR-100 lineage (random init; the encoder never saw the held-out class). Probe and top-1 measure *decodability*; Euclidean AUC, tied Mahalanobis and per-event power at α=0.05 measure *discoverability*. Adding SIGReg is a tie on probe and top-1 and wins every discoverability column – at 100 classes; see Table 18 for the few-class lineage. *Coverage:* 8/8 objectives present; single holdout (class 4), one seed per cell, probe averaged over 3 probe restarts.

5.6 Where the label-free cut does not reach

Table 7 is a negative result about our own method. At a 1% novel rate on clean CIFAR-100 banks, the label-free rule of Section 3.3 engages on 3 of 20 spaces, and where it engages the resulting purity is still below the gate. The spaces where it works share a structure — an NPLM critic on a SIGReg marginal — so the rule should be read as conditional on the scorer, not as a general-purpose procedure. The ceiling algebra says a $20\times$ enrichment is available at this rate; the estimator, not the representation, is what fails to reach it.

6 When it does not work

The rate floor is a signal-efficiency limit, not an information limit. At a 1% novel rate on CIFAR-100 the pipeline as originally built — distance pool, 95th-percentile cut — reaches purity 0.03 on a ceiling of 0.20. The algebra of Section 3.3 says the ceiling is the cut's, and the density-ratio scorer with a derived cut reaches 0.58 with the true labels and 0.39 label-free on the objectives that carry an NPLM critic on a SIGReg marginal. On the leakage-free lineage the label-free rule declines on 17 of 20 spaces at that rate (Table 7); the estimator, not the representation, is what stops at the floor, and we have not fixed it. Three candidate fixes — swapping the estimator, a scale-invariant knee criterion, and injection calibration — all fail on real embeddings, because the bias is not a uniform scale (it grows as the cut widens). Lowering the clusterability constant is the one intervention that helps, and it is a tuning constant we report with its sweep rather than a correction. And the floor is not purely a rate: at the same 1%, flowers is discoverable from the pretrained trunk and aircraft is not (Table 1).

SIGReg does not help by making distance a log-likelihood. Forcing unit class-conditional width on demand is achievable and decorative: the identity is satisfied, and nothing downstream improves. No space in the campaign reaches unit width class-conditionally (widths 0.13–0.79), and seen-class fidelity to the identity *anti*-correlates with novelty calibration. Five independent tests agree that repairing the identity is not the same as improving the space. The defensible claim is the one made in Section 3: SIGReg earns its place empirically, and it is what licenses the unit-variance clustering criterion.

The temperature basin is a landscape finding. At 100 classes a high softmax temperature with the marginal on lifts tied-Mahalanobis from 0.24 to ~ 0.51 and turns per-event detection on. No label-free or transductive criterion finds it — five legal selectors all pick the low temperature — it does not exist at 10 classes (the probe collapses to 0.65), and it costs 0.15–0.31 of supervised top-1. It is reported as a property of the loss landscape, not as a recipe.

The loop repairs weak spaces and damages strong ones. On the leakage-free lineages the fine-tuning loop is probe-positive on every objective whose pre-probe is below 0.90 and probe-negative on those above 0.93, and it pulls every space toward one middle geometry: on CIFAR-100 it raises

objective	DTD / DINO			DTD / LeJEPA		
	mean	sd	≥gate	mean	sd	≥gate
SimCLR	0.005	0.012	0/5	0.002	0.003	0/5
SimCLR+SIGReg	0.003	0.003	0/5	0.021	0.021	0/5
NPLM-bilinear	0.004	0.006	0/5	0.033	0.018	0/5
SupCon	0.028	0.014	0/5	0.073	0.086	1/5
SupCon+SIGReg (λ=5)	0.219	0.016	5/5	0.225	0.007	5/5
NPLM-dist. (sup.)	0.051	0.031	0/5	0.215	0.027	5/5

Table 4: Pool purity is stable across holdout draws, and only one objective is stable across *backbones*. Mean ± sd over draws, and the number of draws clearing the gate. SupCon+SIGReg clears $5/5$ on both backbones; supervised distance-NPLM clears $5/5$ on one and $0/5$ on the other. *Coverage:* DTD, 5 holdout draws per base, single holdout; 12/12 (arm, base) pairs present. Gate = 0.15. Purity read from the run logs.

round	≥gate	95% CI	min	median	mean	$\mathrm{sd}_{\mathrm{seed}}$	$\mathrm{sd}_{\mathrm{draw}}$
r1	143/144	[–, –]	0.103	0.580	–	–	–
r2	143/144	[–, –]	0.103	0.580	–	–	–

Table 5: The frozen-anchor result is reproducible, and the variance that matters is the holdout draw, not the seed. Clopper–Pearson 95% interval on the fraction of cells above the 0.15 gate. Seed-to-seed spread is $\sim 1/8$ of draw-to-draw spread, so single-seed purities elsewhere in this paper carry a draw caveat and not a seed caveat. *Coverage:* galaxy10, 3 backbones × 3 objectives × 5 draws × 3 seeds (+1 archived) = 144 cells; variance stratified within (cell, objective), 9 strata, seed term measured.

SupCon's Euclidean AUC ($0.23 \rightarrow 0.37$) and lowers SupCon+SIGReg's ($0.62 \rightarrow 0.45$) at pool purities of 0.00–0.02. A pool that pure cannot be what drives this; the fine-tune itself is a calibration repair, and it runs whether or not the pool contains anything.

The NP loss transfers as a form, not as a target. Optimising the representation *through* a fitted NP kernel destroys the separation it measures, and warm-starting the kernel at class centroids is harmful: the centroids sit where the density ratio is one. The three roles of Section 3.2 share a functional form and a loss; they do not share learned content.

No-cost is a dataset-dependent claim. SupCon+SIGReg ties SupCon on supervised top-1 on both leakage-free CIFAR lineages (-0.011 and $+0.001$) and loses 0.035 to it on every one of ten galaxy10 draws (Table 40). The residual concatenation ties its parent everywhere. The construction is no-cost; the objective is no-cost except on the one dataset the backbones had never seen.

A Detailed results

This appendix reports the campaign in more detail than the main text, and reports it *as it stands* rather than waiting for completion. All tables in this paper are generated from the archived result files by a single script, and each carries a machine-written coverage line stating what is present and what is missing. Blanks are shown as "–".

galaxy10, and the procedure that clears the gate. galaxy10 carries more weight than any other dataset here, because no backbone we use has seen it: DINO, LeJEPA and VISReg were all pre-trained on natural images, and galaxy morphology is genuinely out of that distribution. It is also, for the same reason, the hardest cell in the study.

Table 8 reports it under both discovery procedures on the same holdout draws. Under natural discovery — a distance pool with the encoder fine-tuned — round-1 pool purity sits at 0.09–0.29 across every objective and backbone, with sds that are frequently as large as the means: almost nothing clears the 0.15 gate, and no objective clears it reliably. Under the frozen-anchor density-ratio pool, the same objectives on the same draws reach 0.39–0.58.

That is a factor of three to five from changing the procedure and nothing else. It is the sharpest

cell	ctor	variant	A	B	C	D	disc	ctor	Δ_{int}
cars_visreg	res	residual	0.223	0.171	0.073	0.066	-0.052	-0.150	+0.045
cars_visreg	res	concat	0.223	0.171	0.228	0.174	-0.052	+0.005	-0.002
cars_visreg	res-nplm	residual	0.223	0.171	0.256	0.284	-0.052	+0.033	+0.080
cars_visreg	res-nplm	concat	0.223	0.171	0.263	0.282	-0.052	+0.040	+0.070
dtd_dino	res	residual	0.040	0.068	0.022	0.253	+0.028	-0.018	**+0.203**
dtd_dino	res	concat	0.040	0.068	0.045	0.235	+0.028	+0.005	**+0.163**
dtd_dino	res-nplm	residual	0.040	0.068	0.113	0.087	+0.028	+0.073	-0.053
dtd_dino	res-nplm	concat	0.040	0.068	0.077	0.130	+0.028	+0.037	+0.025
galaxy10_dino	res	residual	0.021	0.037	0.018	0.014	+0.015	-0.003	-0.020
galaxy10_dino	res	concat	0.021	0.037	0.021	0.018	+0.015	-0.001	-0.018
galaxy10_dino	res-nplm	residual	0.021	0.037	0.281	0.299	+0.015	+0.259	+0.003
galaxy10_dino	res-nplm	concat	0.021	0.037	0.091	0.142	+0.015	+0.069	+0.036
galaxy10_lejepa	res	residual	0.007	0.196	0.001	0.002	+0.190	-0.005	-0.189
galaxy10_lejepa	res	concat	0.007	0.196	0.001	0.012	+0.190	-0.005	-0.179
galaxy10_lejepa	res-nplm	residual	0.007	0.196	0.009	0.021	+0.190	+0.002	-0.177
galaxy10_lejepa	res-nplm	concat	0.007	0.196	0.008	0.206	+0.190	+0.001	+0.008
galaxy10_visreg	res	residual	0.311	0.419	0.243	0.457	+0.108	-0.068	**+0.107**
galaxy10_visreg	res	concat	0.311	0.419	0.295	0.466	+0.108	-0.015	+0.063
galaxy10_visreg	res-nplm	residual	0.311	0.419	0.333	0.471	+0.108	+0.022	+0.030
galaxy10_visreg	res-nplm	concat	0.311	0.419	0.353	0.446	+0.108	+0.043	-0.015

Table 6: Discovery $\times$ construction, per-event power at α=0.05. A parent, B +discovery, C +construction, D both; $\Delta_{\text{int}} = (D-C)-(B-A)$ is zero for an additive stack. Bold marks $\Delta_{\text{int}} > 0.10$. The two largest *totals* (galaxy10) are main effects, not interactions. *Coverage:* 5 cells $\times$ 2 constructions $\times$ 2 variants; 20 complete, 0 lacking a pre-discovery child. One seed per cell.

evidence in the paper for a claim we make throughout: failures that present as properties of the representation are frequently properties of the pipeline wrapped around it. On galaxy10 specifically, the ordering of objectives under natural discovery is close to noise — the sds overlap almost completely — and the ordering under the frozen pool is both tighter and higher. We would not have found this by improving the encoder.

Note the coverage line: the frozen-pool arm of the study covers the three supervised objectives only, so nine of the eighteen cells in that half are absent rather than negative.

The per-dataset appendices that follow (B–D) put every measurement in one place. Each opens with a *summary grid*: one row per (objective, construction) with construction in {plain, +residual concat, +NPLM-residual concat}, and one column pair per metric for the space *before* and *after* the discovery loop — part (A) the probe and per-event geometry, part (B) the dataset-level SparKer, Mahalanobis and MMD power at f=0.02. A dash is a cell that was never measured, so the grids are also the coverage map of the campaign. The detail tables that follow each grid carry the remaining columns (accuracy, LID, pools, draws). This section keeps the cross-cutting tables.

Constructions on the leakage-free trunks. Table 9 gives every construction on the from-scratch CIFAR-100 parents. The pattern behind the summary in Section 4 is visible directly: the bare residual usually costs probe accuracy, the concatenation recovers it, and the detection gain survives in both. The CIFAR-10 counterpart is Table 21; the width control and the after-discovery construction are Tables 13 and 14.

Multi-seed cells. Most cells in this campaign are single-seed, for cost reasons. Table 10 reports one of the exceptions, three seeds on the CIFAR-10 champions. Seed spread there is small — ± 0.001 on the best post-discovery probe and ± 0.002 on its Euclidean AUC — which is consistent with the much larger variance decomposition of Table 5. Note also that the discovery loop *reduces* the probe for `supcon+nplm` ($0.954 \to 0.930$) in all three seeds, which is the seeded version of a pattern we observe throughout: the loop repairs weak spaces and damages strong ones.

Known gaps. At the time of writing: the 2×2 of Table 6 is single-seed throughout, and its one large interaction rests on a single cell; the multi-holdout regime is run only on the label-rich datasets; the leakage-free CIFAR lineages are one holdout each (the draw-resampled shortlist is running); flowers, cars and aircraft have no single-holdout fine-tuned parents, so on those datasets

space	$\hat{b}/b$ (tv)	$\hat{b}/b$ (mass)	rule fires	$q^\star$
nplm_cifar100	0.01	0.01	no	–
nplmcw_cifar100	1.04	0.49	yes	0.0277
nplmsd-res-nplm_(concat)_cifar100	0.15	0.08	no	–
nplmsd-res-nplm_(residual)_cifar100	0.07	0.04	no	–
nplmsd-res_(concat)_cifar100	0.17	0.14	no	–
nplmsd-res_(residual)_cifar100	0.37	0.28	no	–
nplmsd_cifar100	0.92	0.07	no	–
simclr_cifar100	0.09	0.09	no	–
ssig-res-nplm_(concat)_cifar100	0.18	0.04	no	–
ssig-res-nplm_(residual)_cifar100	0.43	0.36	no	–
ssig-res_(concat)_cifar100	0.04	0.03	no	–
ssig-res_(residual)_cifar100	0.01	0.01	no	–
ssig_cifar100	0.36	0.20	no	–
supcon-res-nplm_(concat)_cifar100	0.78	0.21	yes	0.0099
supcon-res-nplm_(residual)_cifar100	0.69	0.55	yes	0.0183
supcon-res_(concat)_cifar100	0.03	0.03	no	–
supcon-res_(residual)_cifar100	0.14	0.07	no	–
supcon_cifar100	0.00	0.01	no	–
supsig_cifar100	0.07	0.04	no	–
visreg_cifar100	0.47	0.40	no	–

Table 7: The label-free cut is objective-specific, not universal. Ratio of estimated to true novel base rate at b=0.01; the rule engages only where an NPLM critic sits on a SIGReg marginal, and even then the resulting purity stays below the gate. Reported as a limit of the rule, not of the representations. *Coverage:* 20 spaces, clean CIFAR-100 banks, b_{true}=0.01; the rule engages on 3, and the best estimator is within $5\times$ on 8.

the training-free construction of Tables 1 and 15 is the only single-holdout evidence (it is also the strongest); and the frozen-space construction of Section 4.3 is quoted from evaluation-mode runs after a BatchNorm fix invalidated the earlier archive, so its older numbers do not reproduce and are not used.

objective	natural discovery			frozen-anchor pool		
	DINO	LeJEPA	VISReg	DINO	LeJEPA	VISReg
SimCLR	0.104±0.089	0.112±0.100	0.093±0.063	–	–	–
SimCLR+SIGReg	0.100±0.088	0.123±0.168	0.110±0.077	–	–	–
NPLM-bilinear	0.104±0.080	0.102±0.051	0.117±0.092	–	–	–
SupCon	0.103±0.138	0.132±0.085	0.290±0.067	0.465±0.158	0.575±0.185	0.574±0.180
SupCon+SIGReg (λ=5)	0.092±0.074	0.182±0.200	0.146±0.042	0.459±0.143	0.516±0.186	0.554±0.169
NPLM-dist. (sup.)	0.131±0.038	0.190±0.137	0.262±0.106	0.393±0.147	0.569±0.179	0.543±0.185

Table 8: galaxy10, the dataset no backbone has seen, under two discovery procedures. Round-1 pool purity, mean ± sd. Left: natural discovery (distance pool, encoder fine-tuned). Right: the frozen-anchor density-ratio pool. Same dataset, same holdout draws, a five-fold difference in purity — the procedure, not the representation, is what clears the gate here. *Coverage:* galaxy10, 3 backbones; natural discovery over 5 draws (18/18 cells present), frozen-anchor pool over 5 draws × 3 seeds (9/18 present — that arm of the study covers the three supervised objectives only).

space	probe	Δ	eucl	Δ	per-ev
parent: SupCon					
(parent)	0.940	–	0.233	–	0.00
res (residual)	0.869	-0.071	0.607	+0.374	0.05
res (concat)	0.942	+0.002	0.521	+0.288	0.03
res-nplm (residual)	0.882	-0.058	0.291	+0.057	0.00
res-nplm (concat)	0.947	+0.007	0.227	-0.006	0.00
parent: **SupCon+SIGReg** **(λ=5)**					
(parent)	0.937	–	0.620	–	0.07
res (residual)	0.860	-0.077	0.604	-0.017	0.08
res (concat)	0.949	+0.012	0.648	+0.028	0.08
res-nplm (residual)	0.900	-0.037	0.464	-0.156	0.01
res-nplm (concat)	0.904	-0.033	0.575	-0.046	0.04
parent: NPLM-dist. (sup.)					
(parent)	0.820	–	0.408	–	0.05
res (residual)	0.834	+0.014	0.524	+0.116	0.01
res (concat)	0.867	+0.047	0.521	+0.114	0.01
res-nplm (residual)	0.827	+0.006	0.527	+0.119	0.06
res-nplm (concat)	0.849	+0.029	0.516	+0.108	0.04

Table 9: All constructions on the leakage-free CIFAR-100 trunks. Δ columns are against that parent. The bare residual generally loses probe accuracy and the concatenation recovers it; the detection gain is large and nearly free. *Coverage:* 3 parents $\times$ 4 constructions on the from-scratch CIFAR-100 trunks; one seed per cell.

space	probe pre	probe post	eucl	mahaT	n
nplmsup+nplm	0.888±0.010	0.917±0.004	0.686±0.016	0.752±0.025	3
nplmsup→res	0.871±0.015	0.879±0.017	0.448±0.015	0.621±0.045	3
sup+nplm	0.926±0.012	0.979±0.001	0.794±0.009	0.729±0.044	3
sup→res-nplm	0.930±0.012	0.983±0.001	0.834±0.002	0.759±0.012	3
supcon+nplm	0.954±0.002	0.930±0.009	0.766±0.005	0.723±0.016	3

Table 10: Multi-seed CIFAR-10 champions. Mean $\pm$ sd over seeds. Seed spread here is small, consistent with Table 5. *Coverage:* CIFAR-10, 5 spaces, up to 3 seeds; one of the few multi-seed cells in the campaign.

space	probe pre	probe post	eucl	mahaT	SpK post	MMD post	n
100-D (50+50), classwise-λ5, 3 seeds							
nplmsup+nplm	0.931±0.008	0.938±0.004	0.502±0.112	0.434±0.093	0.300±0.075	0.807±0.094	3
nplmsup→res	0.913±0.008	0.926±0.007	0.437±0.004	0.341±0.024	0.147±0.038	0.300±0.145	3
sup+nplm	0.928±0.007	0.926±0.007	0.525±0.016	0.416±0.076	0.227±0.159	0.340±0.102	3
sup→res-nplm	0.929±0.003	0.925±0.010	0.516±0.018	0.385±0.024	0.313±0.232	0.440±0.142	3
supcon+nplm	0.954±0.008	0.951±0.005	0.375±0.044	0.403±0.079	0.333±0.050	0.800±0.059	3
32-D (16+16), 3 seeds							
nplmsup+nplm	0.901±0.009	0.902±0.002	0.563±0.026	0.505±0.011	0.207±0.047	0.667±0.034	3
nplmsup→res	0.839±0.006	0.866±0.009	0.480±0.026	0.402±0.011	0.173±0.034	0.500±0.188	3
sup+nplm	0.897±0.009	0.867±0.006	0.576±0.007	0.414±0.012	0.087±0.025	0.447±0.082	3
sup→res-nplm	0.819±0.024	0.835±0.028	0.516±0.019	0.347±0.016	0.173±0.025	0.500±0.016	3
supcon+nplm	0.941±0.001	0.935±0.004	0.456±0.022	0.445±0.025	0.333±0.168	0.753±0.105	3

Table 11: Multi-seed CIFAR-100 champions (hub lineage). Power columns at $f=0.02$, post-discovery. *Coverage:* CIFAR-100, two exp-59 configurations, 6 result files; hub-pretrained trunk (ablation lineage).

space	probe	acc	eucl	mahaT
cifar10, 32d, 5 paired seeds				
parent	0.939±0.002	0.869±0.007	0.722±0.019	0.667±0.026
res_concat	0.955±0.002	0.856±0.004	0.567±0.020	0.541±0.026
res-nplm_concat	0.945±0.002	0.873±0.007	0.717±0.023	0.593±0.026
cifar100, 100d, 5 paired seeds				
parent	0.947±0.010	0.574±0.005	0.385±0.033	0.374±0.021
res_concat	0.952±0.008	0.534±0.004	0.607±0.020	0.494±0.013
res-nplm_concat	0.955±0.006	0.576±0.003	0.339±0.035	0.307±0.014

Table 12: Hub-lineage residual concats, 5 paired seeds (exp 75; ablation lineage). *Coverage:* exp 75, both CIFAR datasets, 5 seeds; hub-pretrained trunk (saw the held-out class).

cell	metric	parent (100)	control (200)	concat (100+100)	concat−control	verdict
cars-visreg	probe	0.707	0.727±0.014	0.855	+0.116	concat
	eucl	0.715	0.675±0.009	0.747	+0.080	concat
	mahaT	0.661	0.591±0.018	0.678	+0.107	concat
galaxy10-dino	probe	0.938	0.944	0.955	+0.011	TIE
	eucl	0.666	0.652	0.559	-0.092	control
	mahaT	0.786	0.777	0.762	-0.014	TIE
galaxy10-lejepa	probe	0.910±0.007	0.917	0.971±0.004	+0.058	concat
	eucl	0.690±0.036	0.734	0.372±0.063	-0.446	control
	mahaT	0.741±0.022	0.799	0.661±0.026	-0.174	control

Table 13: Width-matched control for the residual concat (exp 134a). A same-width non-residual head recovers little of the flagship probe gain and loses geometry on cars/VISReg; on galaxy10 the probe gain is a tie on DINO. *Coverage:* 3 cells; control = the parent objective with a 200-D head (no residual); concat = 100+100 residual concat; TIE unless the gap clears $\max(0.017, \mathrm{sd}_a + \mathrm{sd}_b)$.

Table 14: Residual built after discovery vs. before (exp 134c). The pseudo-labelled child corpus is 100% novel in every cell, including where the pool purity is far below the gate. *Coverage:* 5 cells, archived holdout draw, seed 0; parent = SupCon fine-tune; archived = exp-71 pre-discovery order.

concat	probe (pre)	probe (post)	eucl (pre)	eucl (post)	mahaT (pre)	mahaT (post)	per-ev (pre)	per-ev (post)
cars_visreg: round-1 purity 0.206, 217 pseudo-labelled at purity 1.00								
supcon-ft→res	0.801	0.847	0.705	0.692	0.613	0.607	0.23	0.17
supcon-ft→res-nplm	0.855	0.872	0.747	0.744	0.678	0.675	0.26	0.28
dtd_dino: round-1 purity 0.129, 511 pseudo-labelled at purity 1.00								
supcon-ft→res	0.845	0.898	0.683	0.781	0.604	0.780	0.04	0.23
supcon-ft→res-nplm	0.832	0.864	0.733	0.764	0.648	0.690	0.08	0.13
galaxy10_dino: round-1 purity 0.070, 64 pseudo-labelled at purity 1.00								
supcon-ft→res	–	0.949	–	0.595	–	0.746	–	0.02
supcon-ft→res-nplm	–	0.938	–	0.728	–	0.782	–	0.14
galaxy10_lejepa: round-1 purity 0.059, 9 pseudo-labelled at purity 1.00								
supcon-ft→res	–	0.971	–	0.432	–	0.740	–	0.01
supcon-ft→res-nplm	–	0.943	–	0.739	–	0.797	–	0.21
galaxy10_visreg: round-1 purity 0.481, 108 pseudo-labelled at purity 1.00								
supcon-ft→res	0.964	0.974	0.676	0.807	0.868	0.918	0.30	0.47
supcon-ft→res-nplm	0.946	0.968	0.744	0.820	0.839	0.895	0.35	0.45

Table 15: Frozen-trunk pooling with no training at all, on the pretrained backbone features: distance vs density-ratio pool, multi- vs single-holdout regime. *Coverage:* 90 cell files (multi-holdout default draw, and single-holdout draws 0–4); pretrained ViT-B/16 features, identity head, anchors = seen centroids, no fine-tune; gate $= 0.15$.

	multi-holdout						single holdout (draws)					
	dist			np			dist			np		
cell	r1	r2	$\geq$g	r1	r2	$\geq$g	r1	r2	$\geq$g	r1	r2	$\geq$g
galaxy10/dino	0.036	0.158	0/1	0.615	0.630	1/1	0.164±0.022	0.128±0.002	2/5	0.598±0.061	0.593±0.061	5/5
galaxy10/lejepa	0.024	0.101	0/1	0.585	0.600	1/1	0.182±0.012	0.113±0.022	5/5	0.591±0.040	0.586±0.044	5/5
galaxy10/visreg	0.036	0.059	0/1	0.581	0.577	1/1	0.230±0.012	0.153±0.026	5/5	0.556±0.026	0.574±0.041	5/5
dtd/dino	0.668	0.383	1/1	0.611	0.624	1/1	0.105±0.020	0.074±0.025	0/5	0.257±0.016	0.256±0.027	5/5
dtd/lejepa	0.529	0.351	1/1	0.507	0.507	1/1	0.041±0.024	0.037±0.017	0/5	0.261±0.014	0.246±0.038	5/5
dtd/visreg	0.580	0.295	1/1	0.526	0.552	1/1	0.015±0.014	0.013±0.010	0/5	0.267±0.014	0.245±0.027	5/5
flowers/dino	0.562	0.319	1/1	0.495	0.446	1/1	0.098±0.041	0.051±0.033	1/5	0.158±0.006	0.154±0.010	5/5
flowers/lejepa	0.338	0.287	1/1	0.506	0.521	1/1	0.052±0.025	0.038±0.015	0/5	0.177±0.039	0.150±0.014	4/5
flowers/visreg	0.303	0.193	1/1	0.483	0.575	1/1	0.046±0.038	0.034±0.020	0/5	0.159±0.016	0.160±0.020	3/5
cars/dino	0.070	0.068	0/1	0.104	0.085	0/1	0.003±0.004	0.003±0.002	0/5	0.009±0.008	0.005±0.003	0/5
cars/lejepa	0.054	0.045	0/1	0.074	0.104	0/1	0.004±0.002	0.003±0.002	0/5	0.007±0.008	0.011±0.007	0/5
cars/visreg	0.068	0.061	0/1	0.079	0.119	0/1	0.004±0.004	0.003±0.002	0/5	0.010±0.006	0.005±0.003	0/5
aircraft/dino	0.176	0.130	1/1	0.157	0.437	1/1	0.029±0.018	0.028±0.018	0/5	0.016±0.010	0.021±0.022	0/5
aircraft/lejepa	0.123	0.083	0/1	0.088	0.180	0/1	0.014±0.013	0.017±0.013	0/5	0.010±0.007	0.009±0.008	0/5
aircraft/visreg	0.138	0.085	0/1	0.125	0.208	0/1	0.015±0.012	0.012±0.009	0/5	0.010±0.008	0.009±0.008	0/5

B CIFAR-10 and CIFAR-100, leakage-free lineage: every measurement

All eight objectives of the design cube, trained from random initialisation on the seen classes of each dataset (ResNet-20, 100-D head, 200 epochs, holdout class 4), with the full battery on the frozen space, the frozen pools, the label-free cut, and the fine-tuning loop. CIFAR-10 holds out one class in ten ($b = 0.10$); CIFAR-100 one in a hundred ($b = 0.01$). Every number is seed 0 at one holdout; the draw-resampled shortlist is running at the time of writing. Reading the two datasets side by side is the point: the objective ranking, the discoverability of the frozen space, and the response to the loop all flip between them. In the grids, "post" for a plain space is the fine-tuning loop of exp 68 (encoder trains, distance pool, two rounds); discovery on the scratch residual concats has not been run.

B.1 CIFAR-10, leakage-free

Table 16: CIFAR-10, leakage-free lineage: every (objective, construction) $\times$ (pre, post). (A) Probe and per-event geometry. *Coverage:* 8/8 objectives, from random init, 100-D, holdout 4 (single holdout, $b = 0.10$), seed 0. 'post' = the exp-68 fine-tuning loop (distance pool, 2 rounds). Residual concats: parents SupCon / SupCon+SIGReg / NPLM-dist. only (exp 137); discovery on scratch concats not run.

space	probe		eucl		mahaT		per-event	
	pre	post	pre	post	pre	post	pre	post
SupCon								
plain	0.905	0.901	0.514	0.695	0.552	0.510	0.000	0.003
+res concat	0.939	–	0.571	–	0.567	–	0.001	–
+res-nplm concat	0.916	–	0.576	–	0.585	–	0.000	–
SupCon+SIGReg (λ5)								
plain	0.884	0.898	0.439	0.708	0.454	0.403	0.013	0.233
+res concat	0.932	–	0.407	–	0.383	–	0.016	–
+res-nplm concat	0.919	–	0.571	–	0.544	–	0.054	–
NPLM-dist.+cw SIGReg								
plain	0.921	0.950	0.729	0.789	0.559	0.626	0.194	0.377
+res concat	–	–	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–	–	–
SupCon+SIGReg (repulse)								
plain	0.871	0.876	0.667	0.669	0.582	0.576	0.121	0.219
+res concat	–	–	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–	–	–
NPLM-dist. (sup.)								
plain	0.719	0.908	0.692	0.582	0.659	0.515	0.123	0.097
+res concat	0.876	–	0.378	–	0.459	–	0.017	–
+res-nplm concat	0.842	–	0.610	–	0.543	–	0.030	–
SimCLR								
plain	0.808	0.903	0.227	0.467	0.224	0.340	0.003	0.019
+res concat	–	–	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–	–	–
VISReg/LeJEPA								
plain	0.832	0.881	0.395	0.449	0.377	0.367	0.010	0.035
+res concat	–	–	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–	–	–
NPLM (unsup.)								
plain	0.799	0.843	0.472	0.547	0.468	0.507	0.052	0.106
+res concat	–	–	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–	–	–

Table 17: CIFAR-10, leakage-free lineage: every (objective, construction) $\times$ (pre, post). (B) Dataset-level power at $f = 0.02$, $\alpha = 0.05$. *Coverage:* 8/8 objectives, from random init, 100-D, holdout 4 (single holdout, $b = 0.10$), seed 0. 'post' = the exp-68 fine-tuning loop (distance pool, 2 rounds). Residual concats: parents SupCon / SupCon+SIGReg / NPLM-dist. only (exp 137); discovery on scratch concats not run.

space	SparKer		Mahalanobis		MMD	
	pre	post	pre	post	pre	post
SupCon						
plain	0.88	0.14	0.14	0.40	0.84	0.42
+res concat	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
SupCon+SIGReg (λ5)						
plain	0.16	0.50	0.00	0.04	0.46	0.24
+res concat	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
NPLM-dist.+cw SIGReg						
plain	0.38	0.44	0.20	0.60	0.48	0.48
+res concat	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
SupCon+SIGReg (repulse)						
plain	0.36	0.58	0.56	0.14	0.28	0.36
+res concat	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
NPLM-dist. (sup.)						
plain	0.16	0.02	0.22	0.12	0.16	0.06
+res concat	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
SimCLR						
plain	0.22	0.02	0.00	0.06	0.44	0.12
+res concat	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
VISReg/LeJEPA						
plain	0.04	0.04	0.12	0.20	0.08	0.00
+res concat	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
NPLM (unsup.)						
plain	0.04	0.08	0.02	0.18	0.02	0.00
+res concat	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–

objective	probe	top-1	acc	supAUC	eucl	mahaT	mahaPC	lid	per-ev	probe sd
SupCon	0.905	0.896	0.741	0.915	0.514	0.552	0.646	0.703	0.00	0.002
SupCon+SIGReg (λ=5)	0.884	0.897	0.894	0.989	0.439	0.454	0.390	0.518	0.01	0.001
NPLM-dist.+classwise SIGReg	0.921	0.898	0.896	0.989	0.729	0.559	0.589	0.248	0.19	0.000
SupCon+SIGReg (repulse)	0.871	0.802	0.803	0.963	0.667	0.582	0.649	0.616	0.12	0.000
NPLM-dist. (sup.)	0.719	0.827	0.799	0.969	0.692	0.659	0.646	0.376	0.12	0.022
SimCLR	0.808	0.691	0.638	0.919	0.227	0.224	0.353	0.574	0.00	0.002
VISReg/LeJEPA	0.832	0.697	0.638	0.922	0.395	0.377	0.515	0.621	0.01	0.003
NPLM (unsup.)	0.799	0.485	0.435	0.824	0.472	0.468	0.473	0.484	0.05	0.001

Table 18: CIFAR-10 leakage-free lineage: the frozen-space battery. probe = holdout-vs-rest linear AUC; top-1 = supervised linear probe on seen classes; acc = nearest-centroid; supAUC = macro OvR AUC of the proto posterior; eucl / mahaT / mahaPC = novelty AUC of min-centroid distance, tied and per-class Mahalanobis; lid = local-intrinsic-dimension AUC; per-ev = per-event power at α=0.05. *Coverage:* 8/8 objectives; CIFAR-10 from random init, 100-D, holdout class 4 (single holdout, $b = 0.10$), seed 0; probe and top-1 over 3 probe restarts.

	SparKer			Mahalanobis			MMD		
objective	0.01	0.02	0.1	0.01	0.02	0.1	0.01	0.02	0.1
SupCon	0.48	0.88	1.00	0.16	0.14	0.84	0.32	0.84	1.00
SupCon+SIGReg (λ=5)	0.06	0.16	1.00	0.00	0.00	0.00	0.18	0.46	1.00
NPLM-dist.+classwise SIGReg	0.12	0.38	1.00	0.10	0.20	0.94	0.18	0.48	1.00
SupCon+SIGReg (repulse)	0.08	0.36	1.00	0.24	0.56	1.00	0.08	0.28	1.00
NPLM-dist. (sup.)	0.14	0.16	1.00	0.10	0.22	0.92	0.08	0.16	1.00
SimCLR	0.10	0.22	1.00	0.02	0.00	0.00	0.22	0.44	1.00
VISReg/LeJEPA	0.04	0.04	1.00	0.06	0.12	0.36	0.04	0.08	1.00
NPLM (unsup.)	0.06	0.04	0.88	0.00	0.02	0.02	0.04	0.02	0.92

Table 19: CIFAR-10 leakage-free lineage: dataset-level detection power on the frozen space, before any discovery. Signal fraction f of held-out images injected into a seen-class test sample. *Coverage:* 8/8 objectives; annealed-σ SparKer, parametric Mahalanobis and MMD two-sample power at α=0.05, N_D=5000, 200 null / 50 signal toys; fractions 0.01, 0.02, 0.1.

objective	dist r1	np r1	np r2	np+BN r2	legal$_{10}$	legal$_{30}$	loop r1	loop probe	post eucl	post mahaT
SupCon	0.000	0.547	0.550	0.571	0.731	0.722	0.000	0.904→0.901	0.695	0.510
SupCon+SIGReg (λ=5)	0.029	0.396	0.385	0.441	0.812	0.738	0.029	0.884→0.898	0.708	0.403
NPLM-dist.+classwise SIGReg	0.461	0.485	0.539	0.480	0.757	0.777	0.461	0.922→0.950	0.789	0.626
SupCon+SIGReg (repulse)	0.450	0.354	0.297	0.370	0.380	ref.	0.450	0.873→0.876	0.669	0.576
NPLM-dist. (sup.)	0.285	0.392	0.271	0.448	0.429	0.442	0.285	0.737→0.908	0.582	0.515
SimCLR	0.011	0.443	0.475	0.434	0.848	0.871	0.011	0.804→0.903	0.467	0.340
VISReg/LeJEPA	0.017	0.345	0.290	0.253	0.645	0.583	0.017	0.838→0.881	0.449	0.367
NPLM (unsup.)	0.091	0.264	0.323	0.285	0.376	0.408	0.091	0.799→0.843	0.547	0.507

Table 20: CIFAR-10 leakage-free lineage: pools and discovery. Round-1 purity of the frozen distance and density-ratio (np) pools, the np pool's round 2 with BN frozen and adapting, the label-free cut, and the fine-tuning loop's purity, probe and post geometry. *Coverage:* 8/8 objectives; frozen pools = trunk frozen (BN in eval unless 'BN adapt'), 2×2-epoch anchor rounds; legal cut at $n_{\min} = 10$ and 30 ('ref.' = the rule declined); loop = exp-68 fine-tuning loop, distance pool, 2×5 epochs. Gate $= 0.15$.

space	probe	top-1	acc	eucl	mahaT	lid	per-ev
parent: SupCon							
(parent)	0.905	0.896	0.741	0.514	0.552	0.703	0.00
res (residual)	0.902	0.879	0.864	0.650	0.602	0.622	0.02
res (concat)	0.939	0.899	0.815	0.571	0.567	0.678	0.00
res-nplm (residual)	0.892	0.889	0.872	0.659	0.664	0.421	0.05
res-nplm (concat)	0.916	0.902	0.847	0.576	0.585	0.635	0.00
parent: **SupCon+SIGReg** **(**λ=5**)**							
(parent)	0.884	0.897	0.894	0.439	0.454	0.518	0.01
res (residual)	0.878	0.662	0.583	0.380	0.377	0.551	0.01
res (concat)	0.932	0.896	0.889	0.407	0.383	0.513	0.02
res-nplm (residual)	0.888	0.885	0.872	0.758	0.617	0.566	0.12
res-nplm (concat)	0.919	0.897	0.896	0.571	0.544	0.529	0.05
parent: NPLM-dist. (sup.)							
(parent)	0.719	0.827	0.799	0.692	0.659	0.376	0.12
res (residual)	0.851	0.683	0.606	0.371	0.356	0.552	0.02
res (concat)	0.876	0.835	0.648	0.378	0.459	0.595	0.02
res-nplm (residual)	0.817	0.754	0.560	0.480	0.426	0.653	0.02
res-nplm (concat)	0.842	0.832	0.799	0.610	0.543	0.576	0.03

Table 21: CIFAR-10 leakage-free lineage: every residual construction, with the supervised top-1 that the main text's no-cost claim rests on. *Coverage:* 3 parents $\times$ 4 constructions on the from-scratch CIFAR-10 trunks; seed 0, holdout 4.

B.2 CIFAR-100, leakage-free

Table 22: CIFAR-100, leakage-free lineage: every (objective, construction) $\times$ (pre, post). (A) Probe and per-event geometry. *Coverage:* 8/8 objectives, from random init, 100-D, holdout 4 (single holdout, $b = 0.01$), seed 0. 'post' = the exp-68 fine-tuning loop (distance pool, 2 rounds). Residual concats: parents SupCon / SupCon+SIGReg / NPLM-dist. only (exp 137); discovery on scratch concats not run.

space	probe		eucl		mahaT		per-event	
	pre	post	pre	post	pre	post	pre	post
SupCon								
plain	0.940	0.930	0.233	0.372	0.279	0.396	0.000	0.000
+res concat	0.942	–	0.521	–	0.331	–	0.030	–
+res-nplm concat	0.947	–	0.227	–	0.267	–	0.000	–
SupCon+SIGReg (λ5)								
plain	0.937	0.871	0.620	0.452	0.602	0.514	0.070	0.010
+res concat	0.949	–	0.648	–	0.545	–	0.080	–
+res-nplm concat	0.904	–	0.575	–	0.580	–	0.040	–
NPLM-dist.+cw SIGReg								
plain	0.898	0.937	0.525	0.438	0.411	0.382	0.030	0.020
+res concat	–	–	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–	–	–
SupCon+SIGReg (repulse)								
plain	0.823	0.796	0.340	0.371	0.320	0.365	0.000	0.030
+res concat	–	–	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–	–	–
NPLM-dist. (sup.)								
plain	0.820	0.857	0.408	0.349	0.345	0.310	0.050	0.010
+res concat	0.867	–	0.521	–	0.418	–	0.010	–
+res-nplm concat	0.849	–	0.516	–	0.462	–	0.040	–
SimCLR								
plain	0.799	0.889	0.443	0.315	0.385	0.303	0.010	0.030
+res concat	–	–	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–	–	–
VISReg/LeJEPA								
plain	0.797	0.893	0.499	0.343	0.382	0.293	0.030	0.010
+res concat	–	–	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–	–	–
NPLM (unsup.)								
plain	0.757	0.885	0.413	0.298	0.361	0.270	0.010	0.070
+res concat	–	–	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–	–	–

Table 23: CIFAR-100, leakage-free lineage: every (objective, construction) $\times$ (pre, post). (B) Dataset-level power at $f = 0.02$, $\alpha = 0.05$. *Coverage:* 8/8 objectives, from random init, 100-D, holdout 4 (single holdout, $b = 0.01$), seed 0. 'post' = the exp-68 fine-tuning loop (distance pool, 2 rounds). Residual concats: parents SupCon / SupCon+SIGReg / NPLM-dist. only (exp 137); discovery on scratch concats not run.

space	SparKer		Mahalanobis		MMD	
	pre	post	pre	post	pre	post
SupCon						
plain	0.14	0.02	0.04	0.00	0.50	0.86
+res concat	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
SupCon+SIGReg (λ5)						
plain	0.22	0.08	0.06	0.06	0.70	0.94
+res concat	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
NPLM-dist.+cw SIGReg						
plain	0.20	0.06	0.06	0.00	0.58	0.42
+res concat	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
SupCon+SIGReg (repulse)						
plain	0.14	0.12	0.02	0.02	0.38	0.44
+res concat	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
NPLM-dist. (sup.)						
plain	0.18	0.12	0.04	0.00	0.52	0.82
+res concat	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
SimCLR						
plain	0.06	0.04	0.02	0.00	0.18	0.70
+res concat	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
VISReg/LeJEPA						
plain	0.04	0.10	0.06	0.00	0.08	0.58
+res concat	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
NPLM (unsup.)						
plain	0.00	0.02	0.02	0.00	0.12	0.32
+res concat	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–

objective	probe	top-1	acc	supAUC	eucl	mahaT	mahaPC	lid	per-ev	probe sd
SupCon	0.940	0.608	0.545	0.974	0.233	0.279	0.487	0.740	0.00	0.001
SupCon+SIGReg (λ=5)	0.937	0.597	0.588	0.976	0.620	0.602	0.631	0.734	0.07	0.007
NPLM-dist.+classwise SIGReg	0.898	0.507	0.463	0.946	0.525	0.411	0.553	0.678	0.03	0.000
SupCon+SIGReg (repulse)	0.823	0.360	0.346	0.930	0.340	0.320	0.432	0.537	0.00	0.008
NPLM-dist. (sup.)	0.820	0.254	0.232	0.913	0.408	0.345	0.348	0.542	0.05	0.003
SimCLR	0.799	0.292	0.242	0.859	0.443	0.385	0.453	0.570	0.01	0.012
VISReg/LeJEPA	0.797	0.239	0.190	0.844	0.499	0.382	0.434	0.534	0.03	0.004
NPLM (unsup.)	0.757	0.223	0.181	0.836	0.413	0.361	0.447	0.635	0.01	0.003

Table 24: CIFAR-100 leakage-free lineage: the frozen-space battery. probe = holdout-vs-rest linear AUC; top-1 = supervised linear probe on seen classes; acc = nearest-centroid; supAUC = macro OvR AUC of the proto posterior; eucl / mahaT / mahaPC = novelty AUC of min-centroid distance, tied and per-class Mahalanobis; lid = local-intrinsic-dimension AUC; per-ev = per-event power at α=0.05. *Coverage:* 8/8 objectives; CIFAR-100 from random init, 100-D, holdout class 4 (single holdout, $b = 0.01$), seed 0; probe and top-1 over 3 probe restarts.

objective	SparKer			Mahalanobis			MMD		
	0.01	0.02	0.05	0.01	0.02	0.05	0.01	0.02	0.05
SupCon	0.08	0.14	1.00	0.08	0.04	0.00	0.28	0.50	1.00
SupCon+SIGReg **(**λ=5**)**	0.20	0.22	1.00	0.10	0.06	0.48	0.30	0.70	1.00
NPLM-dist.+classwise SIGReg	0.06	0.20	1.00	0.08	0.06	0.08	0.24	0.58	1.00
SupCon+SIGReg (repulse)	0.08	0.14	1.00	0.02	0.02	0.00	0.20	0.38	1.00
NPLM-dist. (sup.)	0.08	0.18	0.90	0.02	0.04	0.00	0.20	0.52	1.00
SimCLR	0.02	0.06	1.00	0.04	0.02	0.02	0.08	0.18	0.82
VISReg/LeJEPA	0.02	0.04	0.86	0.04	0.06	0.00	0.06	0.08	0.68
NPLM (unsup.)	0.04	0.00	1.00	0.02	0.02	0.00	0.08	0.12	0.94

Table 25: CIFAR-100 leakage-free lineage: dataset-level detection power on the frozen space, before any discovery. Signal fraction f of held-out images injected into a seen-class test sample. *Coverage:* 8/8 objectives; annealed-σ SparKer, parametric Mahalanobis and MMD two-sample power at α=0.05, N_D=5000, 200 null / 50 signal toys; fractions 0.01, 0.02, 0.05.

objective	dist r1	np r1	np r2	np+BN r2	legal$_{10}$	legal$_{30}$	loop r1	loop probe	post eucl	post mahaT
SupCon	0.000	0.052	0.021	0.027	ref.	ref.	0.000	0.939→0.930	0.372	0.396
SupCon+SIGReg (λ=5)	0.008	0.007	0.018	0.018	ref.	ref.	0.008	0.938→0.871	0.452	0.514
NPLM-dist.+classwise SIGReg	0.006	0.031	0.020	0.016	0.048	ref.	0.006	0.899→0.937	0.438	0.382
SupCon+SIGReg (repulse)	0.009	0.024	0.037	0.040	ref.	ref.	0.009	0.838→0.796	0.371	0.365
NPLM-dist. (sup.)	0.008	0.004	0.001	0.020	ref.	ref.	0.008	0.819→0.857	0.349	0.310
SimCLR	0.004	0.024	0.014	0.005	ref.	ref.	0.004	0.819→0.889	0.315	0.303
VISReg/LeJEPA	0.005	0.022	0.015	0.018	ref.	ref.	0.005	0.808→0.893	0.343	0.293
NPLM (unsup.)	0.009	0.022	0.014	0.016	ref.	ref.	0.009	0.757→0.885	0.298	0.270

Table 26: CIFAR-100 leakage-free lineage: pools and discovery. Round-1 purity of the frozen distance and density-ratio (np) pools, the np pool's round 2 with BN frozen and adapting, the label-free cut, and the fine-tuning loop's purity, probe and post geometry. *Coverage:* 8/8 objectives; frozen pools = trunk frozen (BN in eval unless 'BN adapt'), 2×2-epoch anchor rounds; legal cut at $n_{\min} = 10$ and 30 ('ref.' = the rule declined); loop = exp-68 fine-tuning loop, distance pool, 2×5 epochs. Gate $= 0.15$.

space	probe	top-1	acc	eucl	mahaT	lid	per-ev
parent: SupCon							
(parent)	0.940	0.608	0.545	0.233	0.279	0.740	0.00
res (residual)	0.869	0.344	0.248	0.607	0.474	0.630	0.05
res (concat)	0.942	0.604	0.479	0.521	0.331	0.650	0.03
res-nplm (residual)	0.882	0.443	0.346	0.291	0.355	0.638	0.00
res-nplm (concat)	0.947	0.602	0.550	0.227	0.267	0.731	0.00
parent: **SupCon+SIGReg** **(**λ=5**)**							
(parent)	0.937	0.597	0.588	0.620	0.602	0.734	0.07
res (residual)	0.860	0.248	0.178	0.604	0.473	0.597	0.08
res (concat)	0.949	0.595	0.575	0.648	0.545	0.729	0.08
res-nplm (residual)	0.900	0.575	0.531	0.464	0.484	0.716	0.01
res-nplm (concat)	0.904	0.595	0.591	0.575	0.580	0.739	0.04
parent: NPLM-dist. (sup.)							
(parent)	0.820	0.254	0.232	0.408	0.345	0.542	0.05
res (residual)	0.834	0.266	0.221	0.524	0.449	0.567	0.01
res (concat)	0.867	0.365	0.236	0.521	0.418	0.562	0.01
res-nplm (residual)	0.827	0.316	0.283	0.527	0.480	0.574	0.06
res-nplm (concat)	0.849	0.391	0.349	0.516	0.462	0.588	0.04

Table 27: CIFAR-100 leakage-free lineage: every residual construction, with the supervised top-1 that the main text's no-cost claim rests on. *Coverage:* 3 parents $\times$ 4 constructions on the from-scratch CIFAR-100 trunks; seed 0, holdout 4.

B.3 CIFAR-10 and CIFAR-100, hub-pretrained (ablation only)

The hub-pretrained CIFAR cells — every number in our earlier technical report — use an encoder that saw the held-out class. We keep them as an ablation, and the grids below make their coverage explicit: standalone arms (exps 50/53) have pre-discovery geometry and power, and post-discovery probe and power from the proto fine-tune (exp 55) or the NPLM fine-tune (exp 58), but their post-discovery *geometry* was never recorded; the $16{+}16$ concats (exp 59) and the residual fine-tunes (exps 73/74) carry both. The multi-seed champions are Tables 10 and 11, the five-seed residual concats Table 12. Their probe records (0.983 ± 0.000 post-discovery on CIFAR-10; 0.951 ± 0.005 on CIFAR-100) reproduce across seeds; their leakage-free counterparts are 0.950 (NPLM-dist.+classwise SIGReg under the loop) and 0.949 (SupCon+SIGReg residual concat).

Table 28: CIFAR-10, hub-pretrained lineage (ablation): every (space, construction) × (pre, post). (A) Probe and per-event geometry. *Coverage:* 11 standalone arms; hub-pretrained trunk (the encoder saw the held-out class – ablation lineage); holdout 4, seed 0; post geometry for standalone arms was never measured (exp 55 recorded probe and power only).

space	probe		eucl		mahaT		per-event	
	pre	post	pre	post	pre	post	pre	post
Standalone objectives, 32-D (exps 50/53; post = exp 55 proto fine-tune)								
SupCon	0.942	–	0.689	–	0.644	–	0.000	–
SupCon+SIGReg (λ1)	0.882	–	0.637	–	0.577	–	0.000	–
NPLM-dist. (sup.)	0.783	0.963	0.726	–	0.723	–	0.200	0.346
same, NPLM fine-tune (exp 58)	–	0.812	–	–	–	–	–	0.572
NPLM-dist.+cw SIGReg	0.872	0.977	0.752	–	0.729	–	0.163	0.316
same, NPLM fine-tune (exp 58)	–	0.836	–	–	–	–	–	0.575
NPLM-bilinear	0.853	0.892	0.510	–	0.494	–	0.057	0.073
NPLM-bil.+cw SIGReg	0.854	–	0.570	–	0.564	–	0.066	–
NPLM-bil.(sup.)+cw SIGReg	0.781	–	0.516	–	0.514	–	0.045	–
NPLM-dist. (unsup.)	0.790	–	0.500	–	0.481	–	0.005	–
SimCLR	0.867	–	0.236	–	0.229	–	0.000	–
SimCLR+SIGReg	0.851	–	0.311	–	0.285	–	0.002	–
LeJEPA	0.863	–	0.408	–	0.371	–	0.007	–
Concatenated / residual spaces, 16+16 (exp 59; post = discovery on the concat)								
sup→res-nplm	0.934	0.984	0.834	–	0.768	–	0.275	0.511
nplmsup→res	0.865	0.896	0.432	–	0.596	–	0.015	0.076
supcon+nplm	0.954	0.920	0.768	–	0.741	–	0.015	0.116
sup+nplm	0.932	0.979	0.791	–	0.771	–	0.232	0.438
nplmsup+nplm	0.887	0.918	0.705	–	0.760	–	0.138	0.214
Residual fine-tune, 32+32 (exps 73/74/80)								
SupCon parent	0.940	–	0.717	–	0.631	–	0.000	–
+res concat	0.954	0.953	0.646	0.722	0.548	0.621	0.039	0.320
+res-nplm concat	0.947	0.953	0.703	0.794	0.565	0.622	0.000	0.167
res child alone	0.859	–	0.473	–	0.466	–	0.026	–
res-nplm child alone	0.895	–	0.542	–	0.531	–	0.002	–
SupCon+SIGReg (λ1) parent	0.856	–	0.692	–	0.528	–	0.000	–
+res concat	0.914	–	0.555	–	0.426	–	0.011	–

Table 29: CIFAR-10, hub-pretrained lineage (ablation): every (space, construction) $\times$ (pre, post). (B) Dataset-level power at $f = 0.02$, $\alpha = 0.05$. *Coverage:* 11 standalone arms; hub-pretrained trunk (the encoder saw the held-out class – ablation lineage); holdout 4, seed 0; post geometry for standalone arms was never measured (exp 55 recorded probe and power only).

space	SparKer		Mahalanobis		MMD	
	pre	post	pre	post	pre	post
Standalone objectives, 32-D (exps 50/53; post = exp 55 proto fine-tune)						
SupCon	0.60	–	0.28	–	0.88	–
SupCon+SIGReg (λ1)	0.64	–	0.10	–	0.74	–
NPLM-dist. (sup.)	0.86	0.04	0.46	0.50	0.28	0.40
same, NPLM fine-tune (exp 58)	–	1.00	–	0.90	–	0.74
NPLM-dist.+cw SIGReg	0.70	0.06	0.32	0.74	0.12	0.34
same, NPLM fine-tune (exp 58)	–	0.92	–	0.56	–	0.72
NPLM-bilinear	0.10	0.04	0.16	0.36	0.06	0.00
NPLM-bil.+cw SIGReg	0.10	–	0.26	–	0.06	–
NPLM-bil.(sup.)+cw SIGReg	0.04	–	0.08	–	0.00	–
NPLM-dist. (unsup.)	0.26	–	0.14	–	0.00	–
SimCLR	0.06	–	0.00	–	0.36	–
SimCLR+SIGReg	0.06	–	0.00	–	0.14	–
LeJEPA	0.16	–	0.10	–	0.04	–
Concatenated / residual spaces, 16+16 (exp 59; post = discovery on the concat)						
sup$\rightarrow$res-nplm	0.88	1.00	0.70	0.90	0.34	0.90
nplmsup$\rightarrow$res	0.04	0.08	0.30	0.36	0.00	0.02
supcon+nplm	0.26	0.30	0.64	0.88	0.66	0.06
sup+nplm	0.42	0.94	0.84	0.86	0.24	0.80
nplmsup+nplm	0.12	0.04	0.66	0.62	0.08	0.10
Residual fine-tune, 32+32 (exps 73/74/80)						
SupCon parent	0.98	–	–	–	–	–
+res concat	0.26	0.22	–	0.46	–	0.18
+res-nplm concat	0.92	0.58	–	0.80	–	0.42
res child alone	0.00	–	–	–	–	–
res-nplm child alone	0.06	–	–	–	–	–
SupCon+SIGReg (λ1) parent	–	–	–	–	–	–
+res concat	–	–	–	–	–	–

Table 30: CIFAR-100, hub-pretrained lineage (ablation): every (space, construction) × (pre, post). (A) Probe and per-event geometry. *Coverage:* 11 standalone arms; hub-pretrained trunk (the encoder saw the held-out class – ablation lineage); holdout 4, seed 0; post geometry for standalone arms was never measured (exp 55 recorded probe and power only).

space	probe		eucl		mahaT		per-event	
	pre	post	pre	post	pre	post	pre	post
Standalone objectives, 32-D (exps 50/53; post = exp 55 proto fine-tune)								
SupCon	0.942	–	0.332	–	0.341	–	0.000	–
SupCon+SIGReg (λ1)	0.883	–	0.436	–	0.443	–	0.000	–
NPLM-dist. (sup.)	0.803	0.837	0.369	–	0.269	–	0.020	0.010
NPLM-dist.+cw SIGReg	0.886	0.886	0.452	–	0.372	–	0.020	0.020
NPLM-bilinear	0.935	0.891	0.532	–	0.484	–	0.070	0.050
NPLM-bil.+cw SIGReg	0.872	–	0.458	–	0.388	–	0.030	–
NPLM-bil.(sup.)+cw SIGReg	0.925	–	0.377	–	0.316	–	0.010	–
NPLM-dist. (unsup.)	0.737	–	0.477	–	0.409	–	0.010	–
SimCLR	0.854	–	0.395	–	0.336	–	0.010	–
SimCLR+SIGReg	0.863	–	0.519	–	0.458	–	0.040	–
LeJEPA	0.868	–	0.473	–	0.403	–	0.010	–
Concatenated / residual spaces, 16+16 (exp 59; post = discovery on the concat)								
sup→res-nplm	0.798	0.800	0.490	–	0.325	–	0.000	0.020
nplmsup→res	0.845	0.866	0.516	–	0.417	–	0.010	0.000
supcon+nplm	0.942	0.932	0.436	–	0.415	–	0.000	0.000
sup+nplm	0.896	0.860	0.567	–	0.417	–	0.020	0.090
nplmsup+nplm	0.896	0.900	0.526	–	0.489	–	0.030	0.000
Residual fine-tune, 100+100 (exps 73/74/80)								
SupCon parent	0.958	–	0.338	–	0.347	–	0.000	–
+res concat	0.959	0.957	0.575	0.562	0.440	0.483	0.020	0.030
+res-nplm concat	0.957	0.950	0.296	0.530	0.312	0.455	0.000	0.020
res child alone	0.839	–	0.650	–	0.558	–	0.080	–
res-nplm child alone	0.910	–	0.317	–	0.318	–	0.000	–
SupCon+SIGReg (λ1) parent	0.945	–	0.388	–	0.381	–	0.000	–
+res concat	0.942	–	0.568	–	0.479	–	0.040	–

Table 31: CIFAR-100, hub-pretrained lineage (ablation): every (space, construction) $\times$ (pre, post). (B) Dataset-level power at $f = 0.02$, $\alpha = 0.05$. *Coverage:* 11 standalone arms; hub-pretrained trunk (the encoder saw the held-out class – ablation lineage); holdout 4, seed 0; post geometry for standalone arms was never measured (exp 55 recorded probe and power only).

space	SparKer		Mahalanobis		MMD	
	pre	post	pre	post	pre	post
Standalone objectives, 32-D (exps 50/53; post = exp 55 proto fine-tune)						
SupCon	0.18	–	0.06	–	0.64	–
SupCon+SIGReg (λ1)	0.02	–	0.06	–	0.48	–
NPLM-dist. (sup.)	0.14	0.18	0.00	0.02	0.78	0.68
NPLM-dist.+cw SIGReg	0.04	0.12	0.02	0.04	0.52	0.66
NPLM-bilinear	0.04	0.04	0.04	0.08	0.30	0.58
NPLM-bil.+cw SIGReg	0.06	–	0.00	–	0.22	–
NPLM-bil.(sup.)+cw SIGReg	0.06	–	0.00	–	0.68	–
NPLM-dist. (unsup.)	0.08	–	0.04	–	0.24	–
SimCLR	0.12	–	0.00	–	0.14	–
SimCLR+SIGReg	0.00	–	0.00	–	0.10	–
LeJEPA	0.04	–	0.10	–	0.16	–
Concatenated / residual spaces, 16+16 (exp 59; post = discovery on the concat)						
sup$\rightarrow$res-nplm	0.26	0.18	0.04	0.00	0.30	0.52
nplmsup$\rightarrow$res	0.04	0.22	0.04	0.00	0.12	0.52
supcon+nplm	0.26	0.56	0.00	0.08	0.48	0.90
sup+nplm	0.10	0.06	0.04	0.08	0.30	0.46
nplmsup+nplm	0.16	0.14	0.02	0.00	0.62	0.62
Residual fine-tune, 100+100 (exps 73/74/80)						
SupCon parent	0.20	–	–	–	–	–
+res concat	0.18	0.06	–	0.08	–	0.32
+res-nplm concat	0.30	0.18	–	0.02	–	0.84
res child alone	0.24	–	–	–	–	–
res-nplm child alone	0.08	–	–	–	–	–
SupCon+SIGReg (λ1) parent	–	–	–	–	–	–
+res concat	–	–	–	–	–	–

C galaxy10: every measurement, across holdout draws

galaxy10 is the only dataset genuinely out of distribution for all three backbones, and it carries the most weight. Every cell is single-holdout (one morphology of ten, $b \approx 0.1$) and resampled over five held-out classes $\{2, 3, 4, 5, 6\}$; the archived class-9 draw is reported only as a reference. Table 38 is the fine-tuning battery on all three backbones; Table 39 the residual constructions on the same draws; Table 5 the frozen density-ratio pool over draws and seeds; Table 40 the supervised top-1 that fixes the cost of SIGReg on this dataset.

The summary grids below give every (objective, construction) $\times$ (pre, post) cell per backbone as mean $\pm$ sd across the five draws. The residual concats are built on the SupCon and SupCon+SIGReg parents; discovery on a residual concat (exp 72) was run on the SupCon+res concat only, so the other concats' post columns are dashes. The after-discovery construction of Section 4 (exp 134c) is a different pipeline order and is tabulated in Table 14.

C.1 galaxy10 / DINO

Table 32: galaxy10 / dino: every (objective, construction) × (pre, post), across holdout draws. (A) Probe and per-event geometry. *Coverage:* galaxy10 / dino, single holdout, 5 draws (classes 2,3,4,5,6), mean±sd across draws, one seed per draw; residual concats from exp 71 (5 draws), post-discovery on the SupCon+res concat only (exp 72, 5 draws); concat SparKer pre from the archived draw (exp 80).

space	probe		eucl		mahaT		per-event	
	pre	post	pre	post	pre	post	pre	post
SupCon								
plain	0.879±0.026	0.864±0.029	0.526±0.145	0.548±0.135	0.500±0.135	0.500±0.133	0.063±0.092	0.086±0.088
+res concat	0.903±0.027	0.886±0.024	0.497±0.122	0.506±0.135	0.488±0.138	0.470±0.146	0.059±0.078	0.063±0.077
+res-nplm concat	0.890±0.023	–	0.538±0.140	–	0.494±0.134	–	0.062±0.088	–
SupCon+SIGReg (λ5)								
plain	0.804±0.064	0.852±0.042	0.499±0.104	0.517±0.089	0.482±0.114	0.471±0.087	0.040±0.038	0.040±0.029
+res concat	0.869±0.026	–	0.494±0.089	–	0.486±0.116	–	0.052±0.040	–
+res-nplm concat	–	–	–	–	–	–	–	–
NPLM-dist. (sup.)								
plain	0.775±0.097	0.758±0.132	0.585±0.123	0.517±0.066	0.556±0.129	0.518±0.069	0.105±0.121	0.080±0.053
SimCLR								
plain	0.790±0.064	0.837±0.057	0.460±0.125	0.504±0.107	0.476±0.111	0.496±0.101	0.049±0.046	0.045±0.043
SimCLR+SIGReg								
plain	0.801±0.084	0.844±0.042	0.477±0.069	0.519±0.068	0.508±0.093	0.498±0.057	0.045±0.029	0.039±0.023
NPLM-bilinear								
plain	0.684±0.134	0.808±0.038	0.483±0.063	0.496±0.072	0.450±0.120	0.462±0.119	0.045±0.036	0.066±0.084

Table 33: galaxy10 / dino: every (objective, construction) × (pre, post), across holdout draws. (B) Dataset-level power at $f = 0.02$, $\alpha = 0.05$. *Coverage:* galaxy10 / dino, single holdout, 5 draws (classes 2,3,4,5,6), mean±sd across draws, one seed per draw; residual concats from exp 71 (5 draws), post-discovery on the SupCon+res concat only (exp 72, 5 draws); concat SparKer pre from the archived draw (exp 80).

space	SparKer		Mahalanobis		MMD	
	pre	post	pre	post	pre	post
SupCon						
plain	0.04±0.03	0.10±0.05	0.06±0.04	0.11±0.06	0.08±0.08	0.10±0.07
+res concat	0.04	0.05±0.03	–	0.10±0.06	–	0.13±0.13
+res-nplm concat	0.04	–	–	–	–	–
SupCon+SIGReg (λ5)						
plain	0.06±0.03	0.10±0.05	0.06±0.04	0.10±0.07	0.09±0.06	0.15±0.11
+res concat	0.02	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
NPLM-dist. (sup.)						
plain	0.05±0.05	0.07±0.05	0.06±0.03	0.08±0.05	0.10±0.08	0.12±0.11
SimCLR						
plain	0.05±0.03	0.06±0.05	0.05±0.04	0.08±0.06	0.08±0.06	0.08±0.09
SimCLR+SIGReg						
plain	0.04±0.02	0.04±0.01	0.08±0.06	0.11±0.09	0.09±0.05	0.10±0.10
NPLM-bilinear						
plain	0.10±0.04	0.08±0.06	0.05±0.05	0.08±0.04	0.05±0.05	0.09±0.08

C.2 galaxy10 / LeJEPA

Table 34: galaxy10 / lejepa: every (objective, construction) $\times$ (pre, post), across holdout draws. (A) Probe and per-event geometry. *Coverage:* galaxy10 / lejepa, single holdout, 5 draws (classes 2,3,4,5,6), mean$\pm$sd across draws, one seed per draw; residual concats from exp 71 (5 draws), post-discovery on the SupCon+res concat only (exp 72, 5 draws); concat SparKer pre from the archived draw (exp 80).

space	probe		eucl		mahaT		per-event	
	pre	post	pre	post	pre	post	pre	post
SupCon								
plain	0.892±0.056	0.920±0.051	0.591±0.080	0.679±0.139	0.711±0.115	0.726±0.115	0.047±0.027	0.048±0.040
+res concat	0.945±0.033	0.936±0.034	0.630±0.059	0.628±0.084	0.709±0.079	0.743±0.085	0.124±0.059	0.155±0.108
+res-nplm concat	0.931±0.045	–	0.644±0.080	–	0.699±0.089	–	0.086±0.061	–
SupCon+SIGReg ($\lambda 5$)								
plain	0.762±0.181	0.876±0.064	0.633±0.148	0.669±0.126	0.665±0.163	0.674±0.143	0.134±0.119	0.144±0.110
+res concat	0.855±0.094	–	0.627±0.068	–	0.692±0.118	–	0.093±0.047	–
+res-nplm concat	–	–	–	–	–	–	–	–
NPLM-dist. (sup.)								
plain	0.787±0.073	0.815±0.107	0.591±0.133	0.543±0.109	0.613±0.174	0.601±0.149	0.132±0.095	0.033±0.021
SimCLR								
plain	0.829±0.066	0.830±0.101	0.488±0.086	0.553±0.067	0.574±0.107	0.610±0.054	0.050±0.039	0.097±0.125
SimCLR+SIGReg								
plain	0.836±0.060	0.863±0.049	0.519±0.134	0.548±0.131	0.596±0.118	0.599±0.115	0.065±0.070	0.102±0.125
NPLM-bilinear								
plain	0.808±0.091	0.828±0.101	0.581±0.065	0.593±0.096	0.611±0.097	0.599±0.086	0.075±0.028	0.063±0.023

Table 35: galaxy10 / lejepa: every (objective, construction) $\times$ (pre, post), across holdout draws. (B) Dataset-level power at $f = 0.02$, $\alpha = 0.05$. *Coverage:* galaxy10 / lejepa, single holdout, 5 draws (classes 2,3,4,5,6), mean±sd across draws, one seed per draw; residual concats from exp 71 (5 draws), post-discovery on the SupCon+res concat only (exp 72, 5 draws); concat SparKer pre from the archived draw (exp 80).

space	SparKer		Mahalanobis		MMD	
	pre	post	pre	post	pre	post
SupCon						
plain	0.20±0.22	0.24±0.25	0.13±0.06	0.19±0.20	0.23±0.04	0.26±0.24
+res concat	0.04	0.11±0.11	–	0.21±0.18	–	0.15±0.16
+res-nplm concat	0.14	–	–	–	–	–
SupCon+SIGReg (λ5)						
plain	0.18±0.25	0.21±0.18	0.15±0.04	0.16±0.12	0.16±0.14	0.22±0.19
+res concat	0.14	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
NPLM-dist. (sup.)						
plain	0.18±0.12	0.32±0.16	0.08±0.10	0.21±0.18	0.16±0.08	0.22±0.12
SimCLR						
plain	0.04±0.02	0.07±0.05	0.14±0.06	0.19±0.18	0.04±0.03	0.13±0.09
SimCLR+SIGReg						
plain	0.05±0.02	0.08±0.06	0.08±0.05	0.16±0.11	0.10±0.06	0.09±0.07
NPLM-bilinear						
plain	0.08±0.03	0.08±0.06	0.11±0.12	0.20±0.15	0.07±0.04	0.07±0.07

C.3 galaxy10 / VISReg

Table 36: galaxy10 / visreg: every (objective, construction) × (pre, post), across holdout draws. (A) Probe and per-event geometry. *Coverage:* galaxy10 / visreg, single holdout, 5 draws (classes 2,3,4,5,6), mean±sd across draws, one seed per draw; residual concats from exp 71 (5 draws), post-discovery on the SupCon+res concat only (exp 72, 5 draws); concat SparKer pre from the archived draw (exp 80).

space	probe		eucl		mahaT		per-event	
	pre	post	pre	post	pre	post	pre	post
SupCon								
plain	0.906±0.043	0.907±0.039	0.700±0.076	0.686±0.093	0.686±0.074	0.676±0.103	0.095±0.054	0.105±0.097
+res concat	0.947±0.025	0.922±0.037	0.647±0.065	0.584±0.090	0.658±0.061	0.602±0.046	0.087±0.054	0.045±0.051
+res-nplm concat	0.923±0.036	–	0.709±0.097	–	0.711±0.106	–	0.116±0.088	–
SupCon+SIGReg (λ5)								
plain	0.823±0.108	0.890±0.060	0.615±0.118	0.640±0.116	0.581±0.136	0.608±0.113	0.090±0.096	0.139±0.157
+res concat	0.888±0.052	–	0.576±0.095	–	0.586±0.136	–	0.086±0.062	–
+res-nplm concat	–	–	–	–	–	–	–	–
NPLM-dist. (sup.)								
plain	0.738±0.105	0.855±0.056	0.640±0.112	0.629±0.136	0.628±0.138	0.654±0.136	0.129±0.098	0.033±0.019
SimCLR								
plain	0.847±0.053	0.859±0.051	0.486±0.076	0.561±0.066	0.526±0.065	0.558±0.078	0.045±0.019	0.092±0.093
SimCLR+SIGReg								
plain	0.835±0.040	0.861±0.027	0.515±0.084	0.487±0.082	0.536±0.100	0.552±0.100	0.048±0.028	0.053±0.026
NPLM-bilinear								
plain	0.672±0.089	0.816±0.047	0.538±0.095	0.542±0.096	0.498±0.092	0.511±0.092	0.064±0.067	0.070±0.081

Table 37: galaxy10 / visreg: every (objective, construction) $\times$ (pre, post), across holdout draws. (B) Dataset-level power at $f = 0.02$, $\alpha = 0.05$. *Coverage:* galaxy10 / visreg, single holdout, 5 draws (classes 2,3,4,5,6), mean±sd across draws, one seed per draw; residual concats from exp 71 (5 draws), post-discovery on the SupCon+res concat only (exp 72, 5 draws); concat SparKer pre from the archived draw (exp 80).

space	SparKer		Mahalanobis		MMD	
	pre	post	pre	post	pre	post
SupCon						
plain	0.10±0.08	0.11±0.03	0.10±0.11	0.13±0.10	0.20±0.07	0.16±0.12
+res concat	0.18	0.10±0.11	–	0.10±0.07	–	0.16±0.13
+res-nplm concat	0.46	–	–	–	–	–
SupCon+SIGReg (λ5)						
plain	0.10±0.10	0.12±0.08	0.12±0.14	0.19±0.20	0.16±0.06	0.14±0.08
+res concat	0.08	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
NPLM-dist. (sup.)						
plain	0.18±0.13	0.12±0.05	0.16±0.18	0.19±0.09	0.10±0.06	0.09±0.04
SimCLR						
plain	0.02±0.02	0.09±0.08	0.11±0.04	0.16±0.15	0.03±0.03	0.06±0.04
SimCLR+SIGReg						
plain	0.06±0.03	0.02±0.02	0.04±0.04	0.15±0.14	0.10±0.05	0.14±0.10
NPLM-bilinear						
plain	0.07±0.08	0.06±0.08	0.06±0.06	0.09±0.05	0.08±0.03	0.13±0.15

C.4 galaxy10: supporting tables

Table 38: galaxy10: the single-holdout battery across holdout draws, all three backbones. Pre-discovery probe, geometry and per-event power; round-1 pool purity of the fine-tuning loop and the number of draws clearing the gate. *Coverage:* galaxy10, single holdout, draws [0, 3, 5, 7, 8] (distinct held-out classes), 18 (arm, base) rows; mean $\pm$ sd across draws, one seed per draw; purity read from the run logs; gate $= 0.15$.

objective	probe	probe post	acc	eucl	mahaT	per-ev	purity r1	$\geq$gate
galaxy10 / dino, 5 draws								
SupCon	0.879$\pm$0.026	0.864$\pm$0.029	0.616$\pm$0.013	0.526$\pm$0.145	0.500$\pm$0.135	0.063$\pm$0.092	0.103$\pm$0.138	2/5
SupCon+SIGReg (λ=5)	0.804$\pm$0.064	0.852$\pm$0.042	0.588$\pm$0.015	0.499$\pm$0.104	0.482$\pm$0.114	0.040$\pm$0.038	0.092$\pm$0.074	2/5
NPLM-dist. (sup.)	0.775$\pm$0.097	0.758$\pm$0.132	0.442$\pm$0.024	0.585$\pm$0.123	0.556$\pm$0.129	0.105$\pm$0.121	0.131$\pm$0.038	1/5
SimCLR	0.790$\pm$0.064	0.837$\pm$0.057	0.449$\pm$0.007	0.460$\pm$0.125	0.476$\pm$0.111	0.049$\pm$0.046	0.104$\pm$0.089	2/5
SimCLR+SIGReg	0.801$\pm$0.084	0.844$\pm$0.042	0.440$\pm$0.008	0.477$\pm$0.069	0.508$\pm$0.093	0.045$\pm$0.029	0.100$\pm$0.088	1/5
NPLM-bilinear	0.684$\pm$0.134	0.808$\pm$0.038	0.380$\pm$0.035	0.483$\pm$0.063	0.450$\pm$0.120	0.045$\pm$0.036	0.104$\pm$0.080	3/5
galaxy10 / lejepa, 5 draws								
SupCon	0.892$\pm$0.056	0.920$\pm$0.051	0.753$\pm$0.018	0.591$\pm$0.080	0.711$\pm$0.115	0.047$\pm$0.027	0.132$\pm$0.085	2/5
SupCon+SIGReg (λ=5)	0.762$\pm$0.181	0.876$\pm$0.064	0.696$\pm$0.030	0.633$\pm$0.148	0.665$\pm$0.163	0.134$\pm$0.119	0.182$\pm$0.200	1/5
NPLM-dist. (sup.)	0.787$\pm$0.073	0.815$\pm$0.107	0.627$\pm$0.053	0.591$\pm$0.133	0.613$\pm$0.174	0.132$\pm$0.095	0.190$\pm$0.137	3/5
SimCLR	0.829$\pm$0.066	0.830$\pm$0.101	0.521$\pm$0.009	0.488$\pm$0.086	0.574$\pm$0.107	0.050$\pm$0.039	0.112$\pm$0.100	2/5
SimCLR+SIGReg	0.836$\pm$0.060	0.863$\pm$0.049	0.460$\pm$0.015	0.519$\pm$0.134	0.596$\pm$0.118	0.065$\pm$0.070	0.123$\pm$0.168	1/5
NPLM-bilinear	0.808$\pm$0.091	0.828$\pm$0.101	0.533$\pm$0.024	0.581$\pm$0.065	0.611$\pm$0.097	0.075$\pm$0.028	0.102$\pm$0.051	1/5
galaxy10 / visreg, 5 draws								
SupCon	0.906$\pm$0.043	0.907$\pm$0.039	0.775$\pm$0.019	0.700$\pm$0.076	0.686$\pm$0.074	0.095$\pm$0.054	0.290$\pm$0.067	5/5
SupCon+SIGReg (λ=5)	0.823$\pm$0.108	0.890$\pm$0.060	0.744$\pm$0.017	0.615$\pm$0.118	0.581$\pm$0.136	0.090$\pm$0.096	0.146$\pm$0.042	3/5
NPLM-dist. (sup.)	0.738$\pm$0.105	0.855$\pm$0.056	0.616$\pm$0.042	0.640$\pm$0.112	0.628$\pm$0.138	0.129$\pm$0.098	0.262$\pm$0.106	5/5
SimCLR	0.847$\pm$0.053	0.859$\pm$0.051	0.564$\pm$0.021	0.486$\pm$0.076	0.526$\pm$0.065	0.045$\pm$0.019	0.093$\pm$0.063	1/5
SimCLR+SIGReg	0.835$\pm$0.040	0.861$\pm$0.027	0.437$\pm$0.014	0.515$\pm$0.084	0.536$\pm$0.100	0.048$\pm$0.028	0.110$\pm$0.077	2/5
NPLM-bilinear	0.672$\pm$0.089	0.816$\pm$0.047	0.360$\pm$0.029	0.538$\pm$0.095	0.498$\pm$0.092	0.064$\pm$0.067	0.117$\pm$0.092	2/5

Table 39: galaxy10: residual constructions across holdout draws. The paired probe gain of the concatenation over its own parent is the one galaxy10 effect that survives draw variance. *Coverage:* galaxy10, 3 backbones, single holdout, draws [0, 3, 5, 7, 8]; mean ± sd across draws; last column = paired concat−parent probe gain and wins/draws.

space	probe	acc	eucl	mahaT	per-ev	concat−parent (wins)
galaxy10 / dino, 5 draws						
supcon-ft (parent)	0.879±0.026	0.616±0.013	0.526±0.145	0.500±0.135	0.063±0.092	–
supcon-ft-res (residual)	0.891±0.025	0.526±0.011	0.478±0.107	0.467±0.139	0.055±0.048	–
supcon-ft-res (concat)	0.903±0.027	0.594±0.010	0.497±0.122	0.488±0.138	0.059±0.078	+0.023 (5/5)
supcon-ft-res-nplm (residual)	0.862±0.027	0.598±0.011	0.553±0.115	0.504±0.111	0.067±0.066	–
supcon-ft-res-nplm (concat)	0.890±0.023	0.619±0.012	0.538±0.140	0.494±0.134	0.062±0.088	+0.011 (5/5)
ss-ft (parent)	0.804±0.064	0.588±0.015	0.499±0.104	0.482±0.114	0.040±0.038	–
ss-ft-res (residual)	0.852±0.033	0.522±0.014	0.493±0.084	0.487±0.101	0.056±0.042	–
ss-ft-res (concat)	0.869±0.026	0.568±0.013	0.494±0.089	0.486±0.116	0.052±0.040	+0.065 (5/5)
galaxy10 / lejepa, 5 draws						
supcon-ft (parent)	0.892±0.056	0.753±0.018	0.591±0.080	0.711±0.115	0.047±0.027	–
supcon-ft-res (residual)	0.910±0.045	0.638±0.018	0.623±0.058	0.643±0.090	0.114±0.057	–
supcon-ft-res (concat)	0.945±0.033	0.668±0.016	0.630±0.059	0.709±0.079	0.124±0.059	+0.053 (5/5)
supcon-ft-res-nplm (residual)	0.887±0.049	0.702±0.025	0.596±0.119	0.598±0.105	0.083±0.066	–
supcon-ft-res-nplm (concat)	0.931±0.045	0.760±0.022	0.644±0.080	0.699±0.089	0.086±0.061	+0.039 (5/5)
ss-ft (parent)	0.762±0.181	0.696±0.030	0.633±0.148	0.665±0.163	0.134±0.119	–
ss-ft-res (residual)	0.854±0.075	0.611±0.010	0.521±0.123	0.589±0.132	0.061±0.060	–
ss-ft-res (concat)	0.855±0.094	0.706±0.020	0.627±0.068	0.692±0.118	0.093±0.047	+0.093 (5/5)
galaxy10 / visreg, 5 draws						
supcon-ft (parent)	0.906±0.043	0.775±0.019	0.700±0.076	0.686±0.074	0.095±0.054	–
supcon-ft-res (residual)	0.930±0.036	0.691±0.030	0.528±0.068	0.606±0.052	0.056±0.055	–
supcon-ft-res (concat)	0.947±0.025	0.769±0.020	0.647±0.065	0.658±0.061	0.087±0.054	+0.041 (5/5)
supcon-ft-res-nplm (residual)	0.890±0.053	0.774±0.017	0.685±0.109	0.709±0.109	0.114±0.101	–
supcon-ft-res-nplm (concat)	0.923±0.036	0.781±0.019	0.709±0.097	0.711±0.106	0.116±0.088	+0.017 (5/5)
ss-ft (parent)	0.823±0.108	0.744±0.017	0.615±0.118	0.581±0.136	0.090±0.096	–
ss-ft-res (residual)	0.867±0.054	0.645±0.023	0.498±0.082	0.554±0.102	0.048±0.032	–
ss-ft-res (concat)	0.888±0.052	0.740±0.018	0.576±0.095	0.586±0.136	0.086±0.062	+0.065 (5/5)

Table 40: Supervised top-1 (exp 132) on galaxy10: archived draw and the draw interval. SupCon+SIGReg costs 0.035 against SupCon on every draw; the residual concats tie the parent. *Coverage:* 13 cell files; 3 probe restarts per cell; draws are the exp-125 single-holdout draws.

space	top-1 (archived)	top-1 (draws)	n
galaxy10 / dino			
supcon-ft	0.642	0.666±0.011	5
ss-ft	0.597	0.628±0.013	5
nplm-sup-ft	0.474	0.525±0.023	5
simclr-ft	0.556	0.584±0.010	5
sigreg-ssl-ft	0.533	0.562±0.011	5
nplm-bil-ft	0.404	0.537±0.028	5
supcon-ft-res-cat	0.641	0.663±0.010	2
supcon-ft-resnplm-cat	0.642	0.654	1
ss-ft-res-cat	0.599	0.624±0.009	2
galaxy10 / lejepa			
supcon-ft	0.781	0.790±0.018	5
ss-ft	0.745	0.755±0.024	5
nplm-sup-ft	0.704	0.710±0.029	5
simclr-ft	0.617	0.631±0.018	5
sigreg-ssl-ft	0.543	0.582±0.014	5
nplm-bil-ft	0.579	0.614±0.018	5
supcon-ft-res-cat	0.775	–	0
supcon-ft-resnplm-cat	0.775	–	0
ss-ft-res-cat	0.743	–	0
galaxy10 / visreg			
supcon-ft	0.776	–	0
ss-ft	0.757	–	0
nplm-sup-ft	0.636	–	0
simclr-ft	0.612	–	0
sigreg-ssl-ft	0.506	–	0
nplm-bil-ft	0.480	–	0
supcon-ft-res-cat	0.777	–	0
supcon-ft-resnplm-cat	0.777	–	0
ss-ft-res-cat	0.756	–	0

D DTD: every measurement, across holdout draws

DTD is task-novel to the backbones (textures) and is the dataset on which the discovery workhorse was chosen. Single holdout is one texture class of 47 ($b \approx 0.02$), resampled over classes 8, 46, 38, 15 and 16; Table 41 gives all three backbones. The multi-holdout regime (10 classes held out, $b \approx 0.2$) is the one the earlier report quoted, and its purities (0.80) are not comparable with these (0.22).

Table 41: dtd: the single-holdout battery across holdout draws, all three backbones. Pre-discovery probe, geometry and per-event power; round-1 pool purity of the fine-tuning loop and the number of draws clearing the gate. *Coverage:* dtd, single holdout, draws [0, 1, 3, 4, 5] (distinct held-out classes), 18 (arm, base) rows; mean ± sd across draws, one seed per draw; purity read from the run logs; gate = 0.15.

objective	probe	probe post	acc	eucl	mahaT	per-ev	purity r1	≥gate
dtd / dino, 5 draws								
SupCon	0.883±0.040	0.892±0.070	0.739±0.004	0.732±0.045	0.628±0.077	0.020±0.010	0.028±0.012	0/5
SupCon+SIGReg (λ=5)	0.787±0.036	0.702±0.056	0.762±0.005	0.792±0.019	0.732±0.054	0.230±0.053	0.219±0.014	5/5
NPLM-dist. (sup.)	0.749±0.102	0.861±0.044	0.445±0.018	0.632±0.066	0.635±0.079	0.055±0.043	0.051±0.027	0/5
SimCLR	0.940±0.039	0.906±0.040	0.631±0.004	0.550±0.100	0.564±0.111	0.040±0.080	0.005±0.010	0/5
SimCLR+SIGReg	0.890±0.070	0.834±0.076	0.624±0.010	0.627±0.103	0.606±0.090	0.050±0.042	0.003±0.002	0/5
NPLM-bilinear	0.728±0.113	0.879±0.040	0.340±0.073	0.513±0.075	0.502±0.081	0.020±0.019	0.004±0.005	0/5
dtd / lejepa, 5 draws								
SupCon	0.659±0.144	0.758±0.069	0.728±0.004	0.722±0.070	0.712±0.084	0.110±0.162	0.073±0.077	1/5
SupCon+SIGReg (λ=5)	0.803±0.059	0.692±0.123	0.763±0.005	0.787±0.024	0.780±0.027	0.245±0.073	0.225±0.006	5/5
NPLM-dist. (sup.)	0.708±0.097	0.648±0.158	0.614±0.010	0.811±0.056	0.804±0.035	0.195±0.093	0.215±0.024	5/5
SimCLR	0.938±0.037	0.912±0.045	0.618±0.006	0.563±0.111	0.588±0.113	0.030±0.048	0.002±0.002	0/5
SimCLR+SIGReg	0.909±0.057	0.888±0.039	0.605±0.006	0.656±0.072	0.674±0.108	0.085±0.066	0.021±0.019	0/5
NPLM-bilinear	0.798±0.077	0.873±0.063	0.463±0.071	0.583±0.050	0.648±0.034	0.055±0.010	0.033±0.016	0/5
dtd / visreg, 5 draws								
SupCon	0.774±0.101	0.781±0.100	0.762±0.005	0.781±0.052	0.730±0.088	0.175±0.142	0.184±0.031	4/5
SupCon+SIGReg (λ=5)	0.867±0.028	0.737±0.114	0.788±0.007	0.802±0.049	0.782±0.054	0.285±0.066	0.241±0.025	5/5
NPLM-dist. (sup.)	0.702±0.121	0.656±0.083	0.584±0.010	0.747±0.055	0.757±0.034	0.130±0.048	0.163±0.054	4/5
SimCLR	0.899±0.088	0.904±0.077	0.606±0.003	0.515±0.135	0.523±0.122	0.070±0.073	0.023±0.032	0/5
SimCLR+SIGReg	0.923±0.053	0.911±0.049	0.610±0.004	0.664±0.117	0.664±0.129	0.130±0.087	0.023±0.024	0/5
NPLM-bilinear	0.791±0.093	0.854±0.052	0.458±0.103	0.567±0.111	0.576±0.153	0.040±0.025	0.012±0.006	0/5

E Objective naming

Three objectives collide under similar names at two regulariser strengths, and their numbers must not be pooled.

paper name	code name(s)	objective	λ
SimCLR+SIGReg	`ss`, `simclr_sigreg`	instance / cosine / softmax, unsupervised	1
SupCon+SIGReg (weak)	`supcon_sigreg`	supervised / cosine / softmax, τ=0.1	1
SupCon+SIGReg (strong)	`ss-ft`, `ssig`	same objective as the row above	5
SupCon+SIGReg (repulse)	`supsig`	the proto/repulse recipe; *not* SupCon+SIGReg	–
NPLM-dist. + classwise SIGReg	`nplmcw`, `nplm_dist_sup_cw`	supervised / distance / NPLM, class-wise marginal	5
NPLM-dist. (sup.)	`nplmsd`, `nplm-sup-ft`	supervised / distance / NPLM, global marginal	1

The strong and weak SupCon+SIGReg variants are the same objective at different λ (aux-term magnitudes 0.381 vs. 0.066 on a shared batch); the discovery workhorse is the strong one, and every exp-50-era number is the weak one. `ss` and `ss-ft` share an abbreviation and are different algorithms, one unsupervised and one supervised. The from-scratch arms `ssig` and `nplmsd` compute numerically identical losses to their fine-tune twins `ss-ft` and `nplm-sup-ft`.